Magnetosphere Shielding Model (MSM) Technical Report

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Reference: RADMOD/TR/2025001
Issue/Revision: 1. / Rev 0
Date: April 2025

Administration page

Customer Information	
Document reference	RADMOD/TR/2025001
Project title	Energetic Solar Heavy Ion Environment Models (ESHIEM)
Customer Organisation	ESA – European Space Technology Centre
Customer contact	Dr Piers Jiggens, ESA/ESTEC TEC-EES
Contract number	
Project document reference	
Milestone number	
Date due	

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Record of changes		
Issue	Date	Detail of Changes
1.0	04/2025	First created for public release

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Abstract

As part of ESA Contract 4000107025/12/NL/GLC supported by the Technology Research Programme, RadMod Research Ltd is participating in a consortium developing new solar heavy ion and associated shielding models which may be used by spacecraft engineers to predict the radiation threat to future spacecraft mission. The main objective is to provide a statistical model ideally covering species from alpha particles to uranium ions, and which will become part of an updated Solar Energetic Particle Environment Modelling (SEPEM) System. The other objective of the project is to improve the usability of the SEPEM system, particularly in the data plotting and exchange area. This document provides an detailed description of the development of Magnetosphere Shielding Model (MSM). It includes a brief review of the current state of the magnetosphere shielding calculation tools; discussion of the implementation strategy and descriptions of the calculation algorithms; the software architecture design; detailed description of the software code; examples of the model prediction; and results of the validation and verification activities.

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1 Introduction

1.1 Background

The relative biological effectiveness of the heavy ions means they contribute disproportionately to the equivalent dose in tissue and effective dose in humans. Similarly as atomic number, Z, of the ion increases, the particle unrestricted stopping power scales as Z^2 for a given particle energy per unit mass. This leads to the more intense ionisation tracks which in turn increase the likelihood of SEEs, including catastrophic SEEs. In a previous ESA project, Solar Energetic Particle Environment Modelling (SEPEM) System, a new statistical solar energetic particle model has been developed and implemented as an easy-to-use web based tool. In this system, however, the models and their effects for heavy ions are inadequately addressed. Within the ESA Energetic Solar Heavy Ion Environment Models (ESHIEM) Project, a consortium comprising Kallisto Consultancy, DH Consultancy, RadMod Research and TRAD is developing new solar heavy ion models to allow more accurate prediction of the potential environments from solar energetic particle events and their effects on systems and crews. It also encompasses a substantial re-working of the SEPEM system to improve its usability.

1.2 Purpose

The main objective of this document is to capture the implementation process of the new MSM tool , these include an review of the current state of magnetosphere shielding calculation tools, a description of the implementation strategy and algorithms to be used. The software architecture design and detailed description of the software modules. Other subjects covered are demonstrations of the model prediction and the verification and validation activities.

1.3 Structure and scope

The remaining document comprises the following chapters:

- Chapter 2 provides a review of the current available tools for magnetosphere shielding calculations.
- Chapter 3 discusses the strategy adapted for the implementation and the algorithms which are used in the MSM tool.
- Chapter 4 provides examples of the MSM predictions
- Chapter 5 covers the verification and validation process

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1.5 List of Abbreviations and Acronyms

CL	confidence limit or confidence level
ESA	European Space Agency
ESHIEM	Energetic Solar Heavy Ion Environment Models Project
ESTEC	European Space Technology and Research Centre
GCR	galactic cosmic radiation galactic cosmic ray
Geant4	Geometry and Tracking code version 4
GEO	geostationary Earth orbit
GIOVE	Galileo In-Orbit Validation Element
GTO	geostationary transfer orbit
HEO	Highly elliptical orbit
HI	heavy ion
LET	linear energy transfer
MEO	medium-altitude Earth orbit
MULASSIS	Multi-Layered Shielding Simulation Software
NASA	National Aeronautics and Space Administration
NIEL	non-ionising energy loss
SEE	single event effect
SEP	Solar Energetic Particle
SEPEM	Solar Energetic Particle Environment Modelling
SEU	single event upset
SPENVIS	Space Environment Information System
SR	System Requirement
UR	User Requirement
TID	total ionising dose

1.6 Acknowledgement

We acknowledge numerous valuable discussions Margret Shea and Don Smart concerning rigidity cut-off calculations. They also kindly provided the RCINTUT3 code from which the interpolation algorithm and routine have been adapted into MSM. Also we thank Laurent Desorgher for updating the MAGNETOCOSMICS code which was used for producing the pre-calculated cut-off maps.

2 A Brief Review of Geomagnetic Shielding Calculation Tools

To reach a location within the Earth's magnetosphere, charged solar energetic particles have to penetrate through the shielding offered by the magnetic fields around the Earth which consists of the internal geomagnetic field which is generated by the motion of molten iron alloys in the Earth's outer core and the external magnetospheric fields which are due to variability of the solar wind, with the ring current, the Chapman-Ferraro currents on the magnetopause, the tail current sheet, and the field aligned Birkeland current systems I and II [1][2].

The internal geomagnetic field is generally described in spherical harmonic moments and can be reasonably approximated by a tiled dipole with moment M = $31500r_e^3$ in units of nanotesla (nT), displaced from the Earth's centre by 434 km or $0.068r_e$. The dipole tilt angle is 11° at 79° N and 70°W. The quadruple contribution is about 10% at the surface and decreases to 5% at $2r_e$. The International Geomagnetic Reference Field (IGRF) model [3] represents the most precise model of the geomagnetic field. The Gauss coefficients are derived from magnetic field measurements from geomagnetic stations, ship-towed magnetometers and satellites. Every five years, the International Association of Geomagnetism and Aeronomy (IAGA) issues a new set of Gauss coefficients defining the new IGRF model. The previous IGRF model becomes then a Definitive Geomagnetic Reference Field (DGRF) and the second order terms, defining the secular variation of the field over the past five years, become fixed. In the IGRF model, the Gauss coefficients for a given period are obtained by interpolating and extrapolating the different DGRF/IGRF parameters released every five years and the last version IGRF11 was released in 2010.

The semi-empirical Tsyganenko magnetic field models have been widely utilized in the space physics community for many years. They are best-fit representations for the magnetospheric field, based on a large number of satellite observations. There are currently four main versions. The 1989 version (TYS89) [1] was primarily developed as a simple empirical approximation for the global magnetosphere, binned into several intervals of the disturbance index Kp. The 1996 version (TYS96) [2] has an explicitly defined realistic magnetopause, large-scale Region 1 and 2 Birkeland current systems, and Interplanetary Magnetic Field (IMF) penetration across the boundary. TYS96 is parameterized by the solar wind ram pressure, disturbance storm time index (Dst) , and transverse components (BY and BZ) of the IMF. The 2001 version (TYS01) [4]represents the variable configuration of the inner and near magnetosphere for different interplanetary conditions and ground disturbance levels. It also takes into account the observed dawn-dusk asymmetry of the inner magnetosphere due to the partial ring current that develops during magnetospheric disturbances. The 2005 version (TYS05) [5] is a dynamical model of the storm-time geomagnetic field in the inner magnetosphere, based on space magnetometer data taken during 37 major events in 1996–2000 and concurrent observations of the solar wind and IMF.

The rigidity, R defined as momentum over unit charge, is often used in characterization of the interaction of charged particles with the geomagnetic field. The most commonly used method in determining the transmission of charged particles in the geomagnetic field is the use of the local vertical cut-off rigidity, R_{VC}, for which transmission is unity provided $R > R_{VC}$, and zero otherwise. In reality the vertical cut-off rigidity at a given location inside the magnetosphere is not a simple step function, rather it has a complicated forbidden and transmission penumbra structure between a lower and upper cut-off rigidities. The effective cut-off rigidity is the average between the lower and upper rigidities that accounts for the transparency of the penumbra. From here on we simply refer the effective vertical cut-off rigidity as the vertical cut-off rigidity.

2.1 Stömer Theory

The rigidity in the Earth's magnetosphere can be very crudely estimated using the Störmer theory [6][7] with its dipole moment. For a simple dipole magnetic field, *Rc* at a given direction (ε,φ), can be calculated using the formula:

$$Rc = \frac{C_D \cos^4 \lambda_m}{r_D^2 (1 + \sqrt{1 - \sin\varepsilon \sin\varphi \cos^3 \lambda_m})^2} \tag{1}$$

where λ_m is the magnetic latitude, ε is the zenith angle and ϕ the azimuthal angle measured clockwise from magnetic north; r_D is the distance from the magnetic dipole centre and C_D is a constant directly proportional to the dipole moment. The angular rigidity cut-offs are lowest from the magnetic west and maximum from the magnetic east, while the cut-offs in the magnetic north and south directions are the same as the vertical cut-off. The vertical cut-off rigidity where ε is zero, then

$$R_{VC} = \frac{C_D}{4r_D^2} \cos^4 \lambda_m \tag{2}$$

2.2 MAGNETOCOSMICS

For accurate calculation of the rigidity, it requires the use of cosmic ray tracing method in conjunction with the detailed field models, i.e. the use of the IGRF and Tsyganenko models [8]. Today the most commonly used code for this is MAGNETOCOSMICS [9][10], which was developed by University of Bern with ESA support. It is based on the Geant4 Toolkit [11] and it has been use the production tool for many of the interpolation tools as described next.

2.3 Interpolation Based Tools

Rigidity calculation using the particle trajectory tracing method is very computation intensive and one track could take up to 1 minute of CPU time even on today's fast desktop machines. So instead of calculating the cut-off on-the-fly, pre-calculated vertical cut-off maps are generally used.

The most commonly used rigidity cut-off maps are those produced by Smart et al. [12] for the International Space Station (ISS) orbits. These maps can be scaled to other LEO orbits using the MacIlwain parameter L, as implemented in the RCINTUT3 program.

A similar tool based on the use of cut-off rigidity maps at 100km altitude using the MAGNETOCOSMICS code has been developed and form part of the Models for Atmospheric Ionising Radiation Effects -MAIRE [13][14].

2.4 Other Tools

For LEO, Nymmik *et al.* [15] has derived a fast method for calculating the vertical cut-off rigidity based on the use of a set of pre-calculated cut-off maps at 450 km (using the IGRF model only) and correction function Δ, whose parameters are obtained from the differences between the calculated cut-off maps using IGRF and IGRF+TS89 models.

Another method for calculating the cut-off rigidities has been developed by Badavi [16] in a tool named GEORAD. This tool uses the IGRF model only hence its application is restricted to LEOs. Furthermore GEORAD is under ITAR restrictions thus its availability is limited.

Recently Maget et al. [17] have developed another magnetosphere shielding tool based on the cosmic ray back tracing method, similar to that of the MAGNETOCOSMICS. Instead of Tsyganenko models, it uses the Olson-Pfitzer model for the external field. It is also provides method for obtaining the transmission functions in low earth orbits and beyond.

3 Implementation Strategy and Algorithms

The propagation of SEPs through the Earth's magnetosphere will be calculated in terms of the transmission function for the given date-time and location within the magnetosphere. The transmission function is defined as the product of the average rigidity transmission function and the Earth shadowing factor for the given location and epoch:

$$TR(\lambda_g, \varphi_g, h, t) = SH(h) \times T(R, R_{VC}) \qquad (3)$$

Where λ_g, φ_g and h are the latitude, longitude and altitude of the given location in geographic coordinates and, t the given epoch; R is the rigidity of a SEP and R_{VC} is the vertical cut-off rigidity at the given location and date-time. $T(R, R_{VC})$ is not a simple step function and it is defined in section 3.7, while the *SH(h)* is defined in section 3.8.

The MSM tool will be implemented using the interpolation method. In this section the basis and the algorithms to be used are described in some detail.

3.1 Updating the MAGNETOCOSMICS code

The last release of the MAGNETOCOSMICS code was a few years old now and it was before the IGRF11 become available. Hence it is necessary to update the code for

1. Use of the latest IGRF model, i.e., IGRF11
2. To work with the latest version of Geant4 toolkit, version 10.0

These changes have been committed to the MAGNETOCOSMICS repository in spitfire at ESTEC.

3.2 TSY89 external field model

We selected to use the latest IGRF models, i.e. IGRF11 as the internal field model. We decided to use the TY89 model as the outfield model because it only employs a single parameter Iopt to describe the status of the field. The iopt is related to the better known Kp idex,

Iopt	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Kp	0, 0+	1-,1,1+	2,-2 ,2+	3,-3 ,3+	4,-4, 4+	5-,5 ,5+	6-,6, 6+

The use of a single active level parameter for the external field greatly simplifies the parameterizations of the magnetosphere, as discussed below. However as the original TY89 model only applies to Kp index up to 6, it is necessary to use the Boberg extension [18] to cover the more disturbed conditions with kp > 6. However following discussions with Smart & Shea [19], we decided to use a Dst step size of -50 nT instead of the original Boberg step size of -100 nT. The new extension scheme is :

Kp	**Extended TS89 Model**

	Iopt	Dst (nT)
0, 0+	1	n/a
1-,1, 1+	2	n/a
2-,2, 2+	3	n/a
3,-3 ,3+	4	n/a
4,-4, 4+	5	n/a
5-,5 ,5+	6	n/a
6-,6, 6+	7	n/a
7-,7,7+	7	-50
8-,8,8+	7	-100
9-,9	7	-150

3.3 Parameterization of the magnetosphere for pre-calculated cut-off maps

The pre-calculated rigidity cut-off maps will be generated over the following set of parameters

- Altitude: 450 km.
- Epoch: 1955 to 2015, in 5 year step
- at 0, 3, 6, 9, 12, 15, 18, and 21 UT
- Geomagnetic indices: Kp =0, 1, 2, 3, 4, 5, 6, 7, 8, and 9

3.4 Production of cut-off rigidities maps

A set of 1040 global rigidity cut-off maps with 5° x 5° resolution will be produced, by running the MAGNETOCOSMICS code with the parameters defined in above section. At each grid of the map three key values of the vertical cut-off rigidity: the effective, the lower limit and the upper limit are calculated. It is also necessary to have the corresponding McIlwain "L" parameter and an appropriate magnetic latitude included in the map files. The IRBEM library [20] can be used for calculating both.

3.5 Interpolation algorithms

It has been found [21] [22][23] that the cut-off rigidity change with radial distance is proportional to the inverse power of the McIlwain "L" parameter, i.e. L^{-2}. The "L" parameter can be utilized in the cut-off equation for the cosine squared of the magnetic latitude, hence the vertical cut-off rigidity equation used has the form of:

$$R_{VC} = \frac{V(k)}{L^2} \qquad (3)$$

where *V(k)* is the equivalent of the Störmer constant for the vertical cut-off rigidity. There is no constant global value for *V(k)*, due to the displacement of the effective magnetic centre from the geo-centre and the deviation of the IGRF magnetic field model from a dipole, there is a slightly different *V(k)* at each location.

These characteristics of the “L” parameter form the basis of our basic cut-off rigidity interpolation procedure. With the pre-calculated maps, the cut-off rigidity at a given time and location in the magnetosphere can be determined by interpolation. The pre-calculated map files provide the cut-off values and the "L" value, and with these the constant *V(k)* at every grid point can be obtained.

For any location at the same altitude of the maps, i.e 450 km, we determine the cut-off value at the spacecraft latitude and longitude in the following manner:

1. From the given date determining which two sets of maps are used, e.g., at epochs 2000 and 2005.
2. Determine the Kp value and Universal Time for which the interpolation is to be made.
3. For each set of the maps, do the following
4. Use equation (3) calculating the *V(k)* value for each grid point.
5. Use the geographic coordinates of the location of the spacecraft locate the appropriate “box” for which the cut-off rigidity value is desired.
6. Interpolate in latitude on each side of the box using a weighted average of the change in the “L” coordinate from the top to the bottom of the box. This establishes the constants *V(k)* for the cut-off equation (3) for the desired latitude at the “left” and “right” sides of the box.
7. Employ linear interpolation to determine *V(k)* at the longitude of the spacecraft.
8. Calculate the “L” value for the geographic coordinates of this location.
9. Determine the vertical cut-off rigidity for this location (at 450 km altitude) using the vertical cut-off rigidity equation (3)
10. Employ linear interpolation between the two vertical cut-off rigidities to determine the value for the specified date.

If the spacecraft altitude is not at 450 km, then the McIlwain "L" coordinate is computed at the spacecraft position and altitude and "L" interpolation is used to extrapolate from the cut -off value at 450 km altitude to the spacecraft altitude. The previously computed values of V(k) for the spacecraft latitude and longitude are used with the "L" coordinate of the spacecraft latitude, longitude and altitude to compute the cut off value.

3.6 Cut-off at other directions

The interpolated vertical geomagnetic cut-off rigidity can be transformed to any given direction using the Störmer theory. From equations (1) and (2) we can have the cut-off rigidity, *Rc,* for any given direction (ε, ϕ) derived from the vertical cut-off rigidity, *Rvc:*

$$Rc = \frac{4Rvc}{(1+\sqrt{1-\sin\varepsilon\sin\varphi\cos^3\lambda_m})^2} \qquad (4)$$

Where λ_m is the magnetic latitude of the given location.

Strictly speaking This formula is only applicable to dipole magnetic field, however it is a good enough approximation for near Earth orbits.

3.7 The average transmission function

For charged particles with a rigidity of *R* arriving from interplanetary space at the direction of (ε, ϕ), its transmission function is defined as unity if R > Rc and zero otherwise:

$$T(\varepsilon,\varphi,R) = 1;\ \ R > Rc \qquad (5)$$

$$T(\varepsilon, \varphi, R) = 0; \;\; R \;\leq Rc$$

However the more commonly used parameter for isotropic radiation environment is the averaged transmission function, which is simply the average transmission function over the 4π solid angles:

$$T_{aver}(R) = \iint T(\varepsilon, \varphi, R)\; sin\varepsilon \; d\varepsilon d\varphi / 4\pi \qquad \textbf{(6)}$$

3.8 Earth shadowing

The Earth shadowing effect, *SE*(*h*) can be calculated using the simple geometric formula:

$$SE(h) = 2\pi \times (1 + \frac{\sqrt{(r_e + h)^2 - r_e{}^2}}{r_e + h}) \qquad \textbf{(7)}$$

Where h is the altitude and r_e is the radius of the Earth.

4 Example of the MSM prediction

4.1 Test_orbit.txt

When execute the msm tool with the default "mem.nml" file, it performs calculation using the "test_orbit.txt" orbit file as input. This file contains one day's orbital data of the ISS, with a time resolution of 1 minute. The calculated results are outputted in the file "test_results.txt".

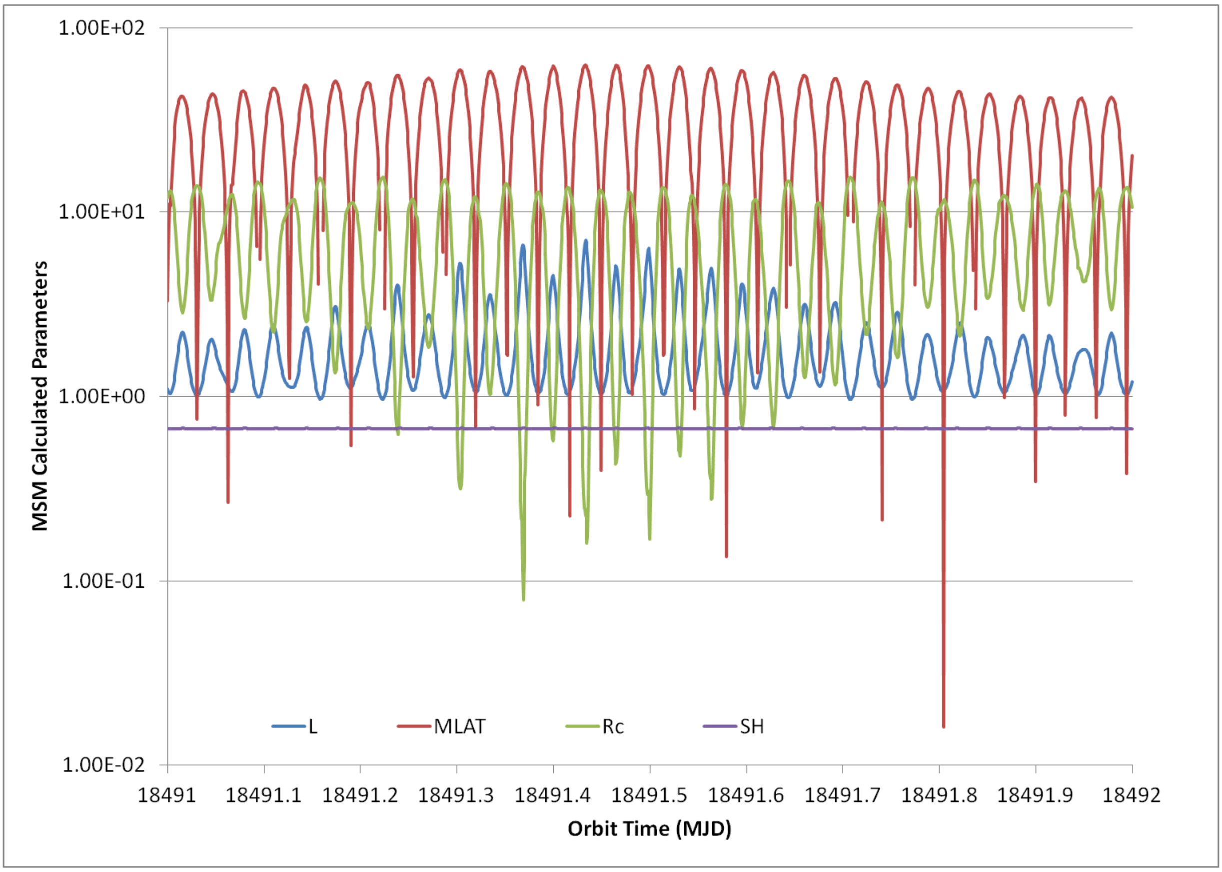

Figure 1: The MSM calculated Mcilwain parameter (L), magnetic latitude (MLAT), vertical cut-off rigidity (Rc) and Earth shadowing factor (SH), for the ISS orbit.

In Figure 6 the calculated Mcilwain parameter (L), magnetic latitude (MLAT), vertical cut-off rigidity (Rc) and Earth shadowing factor (SH) are plotted as against the orbit time. As expected the L and MLAT are correlated and are anti-correlated with the Rc. The SH is almost constant as the orbit altitude is around 400 km.

In Figure 7 we show the calculated transmission function for the first half hour of the orbit file. It covers more than a 1/4 of an orbit hence the highest and lowest transmissions. The transmission function is not a step function of the rigidity. At the small L position, it increases sharply and then flattens out as the corresponding L or Rc increases. At the equator, only > 40 GV particles can reach 100% transmission.

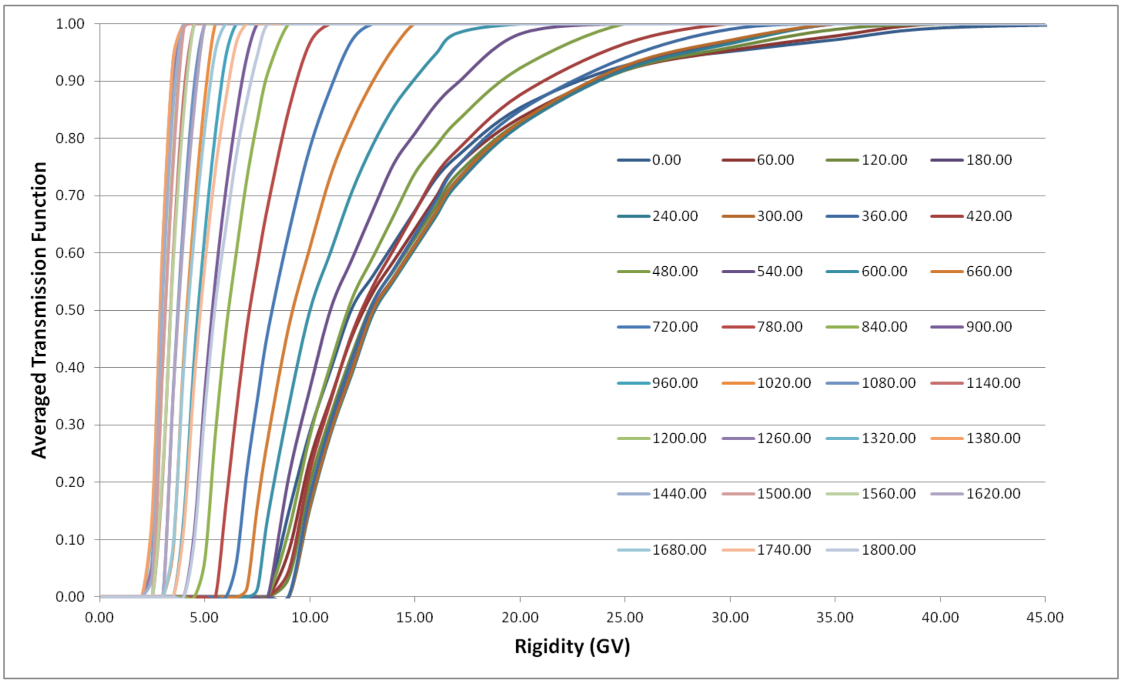

Figure 2: The calculated transmission function for the first half hour orbit positions in the input file. Each line is separated by 1 minute (60 s) in time.

4.2 Global vertical cut-off map

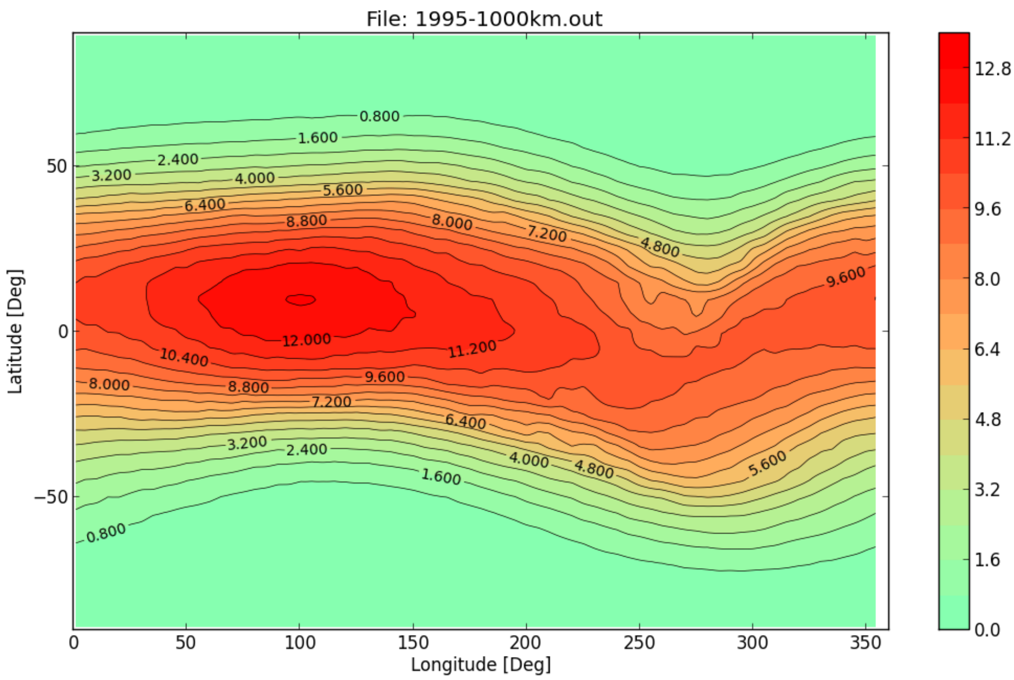

Figure 3: MSM calculated global vertical cut-off map, in GV, at 1000 km altitude, UT =0, 01/01/1995, and Kp = 0 for the geomagnetic field.

The global cut-off map at a given altitude, time/date and Kp condition can be quickly obtained with the MSM tool. Figure 8 shows an example of such a map for the conditions 1000 km altitude, UT =0, 01/01/1995, and Kp = 0.

5 Verifications and Validations

An ad-hoc version of the MSM code "msm/validation/compmaps.for" has been created to produce global vertical cut-off maps which can be direct compared to the maps produced by the MAGNETOCOSMICS and RCINTUT3.

5.1 Against the MAGNETOCOSMICS code

We produce a global map of the difference between the MSM and MAGNETOCOSMICS maps and perform a statistic analysis of the relative of the differences which is defined as

$$Diff\_rela = \frac{(Rc\_msm - Rc\_mgc)}{Rc\_mgc} \qquad (7)$$

In the subsequent sections we show the distributions of the relative difference for various altitudes at 01/01/1995, ut =0, kp=0, and the table below summaries the derived mean and standard deviation of the distribution:

Altitude (km)	**Mean**	**Standard deviation**
20	-0.012	0.0215
100	-0.007	0.0188
500	0.0002	0.015
1000	0.001	0.020
5000	0.05	0.05
10000	0.10	0.23

5.1.1 1995.0, UT = 0, Kp = 0, Alt = 20km

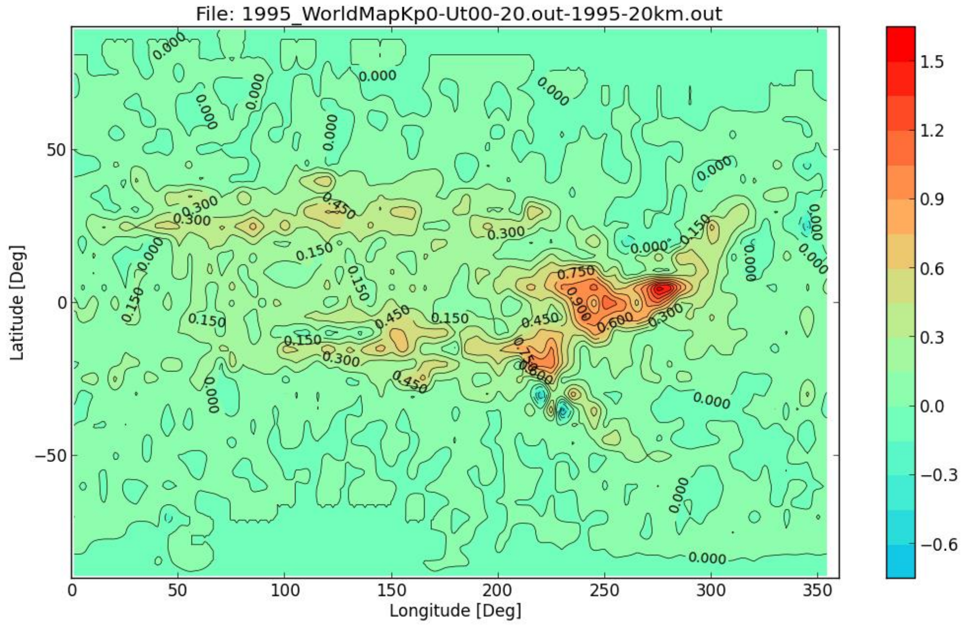

Figure 4: The difference between the MSM and MAGNETOCOSMICS vertical cut-off rigidity (in GV) at global level. There are hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps and in the MAGNETOCOSMICS map.

The difference in the MSM and MAGNETOCOSMICS calculated vertical cut-off rigidity (in GV) are plotted in Figure 9. One can see There are some hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps and in the MAGNETOCOSMICS map. The larger structure patterns are the real difference between the two predictions. In Figure 10 we plotted the distribution of the relative difference for all positions with Rc > 0.5 GV. Assuming they follow the Gaussian distribution, the corresponding mean is -0.012 and the standard deviation is 0.0215

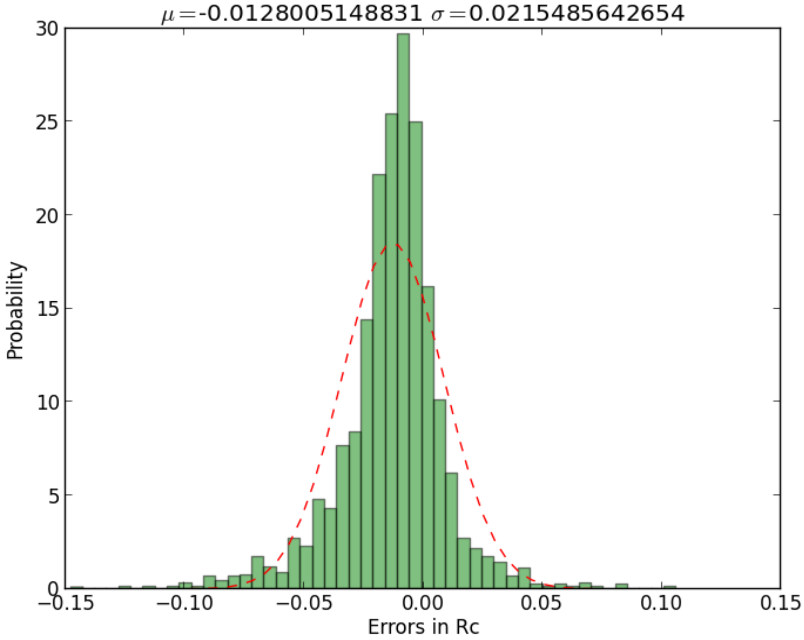

Figure 5: The probability distribution of the differences between the MSM and MAGNETOCOSMICS calculated vertical cut-off rigidity, for locations with Rc > 0.5 GV.

5.1.2 1995.0, UT = 0, Kp = 0, Alt = 100km

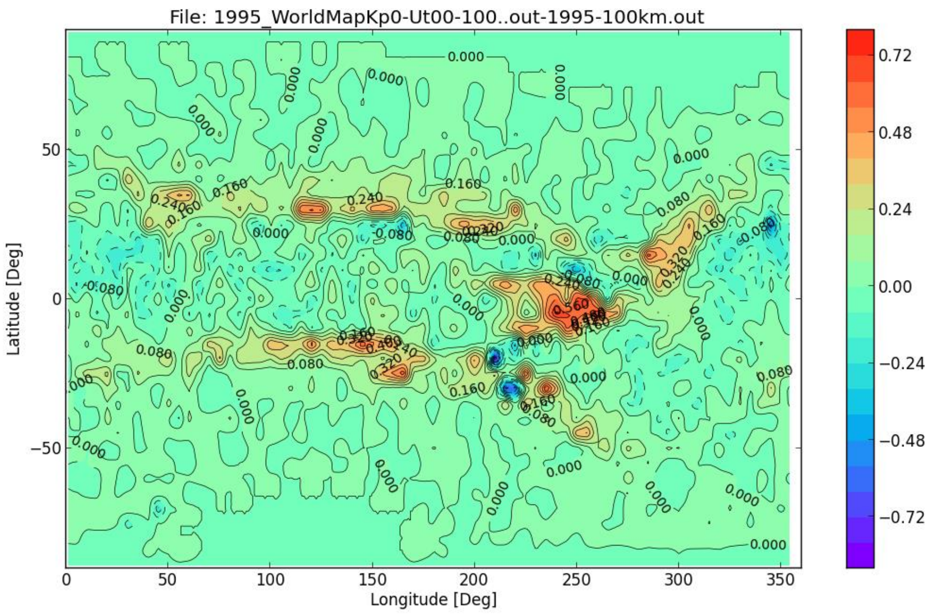

Figure 6: The difference between the MSM and MAGNETOCOSMICS vertical cut-off rigidity (in GV) at global level. There are hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps and in the MAGNETOCOSMICS map.

The difference in the MSM and MAGNETOCOSMICS calculated vertical cut-off rigidity (in GV) are plotted in Figure 11. One can see There are some hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps and in the MAGNETOCOSMICS map. The larger structure patterns are the real difference between the two predictions. In Figure 12 we plotted the distribution of the relative difference for all positions with Rc > 0.5 GV. Assuming they follow the Gaussian distribution, the corresponding mean is -0.007 and the standard deviation is 0.0188

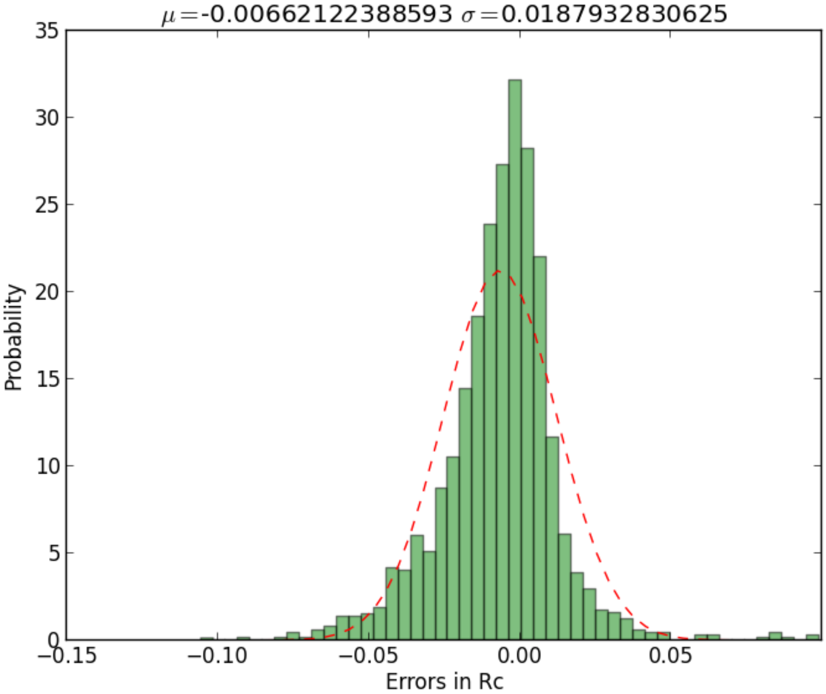

Figure 7: The probability distribution of the differences between the MSM and MAGNETOCOSMICS calculated vertical cut-off rigidity, for locations with Rc > 0.5 GV.

5.1.3 1995.0, UT = 0, Kp = 0, Alt = 500km

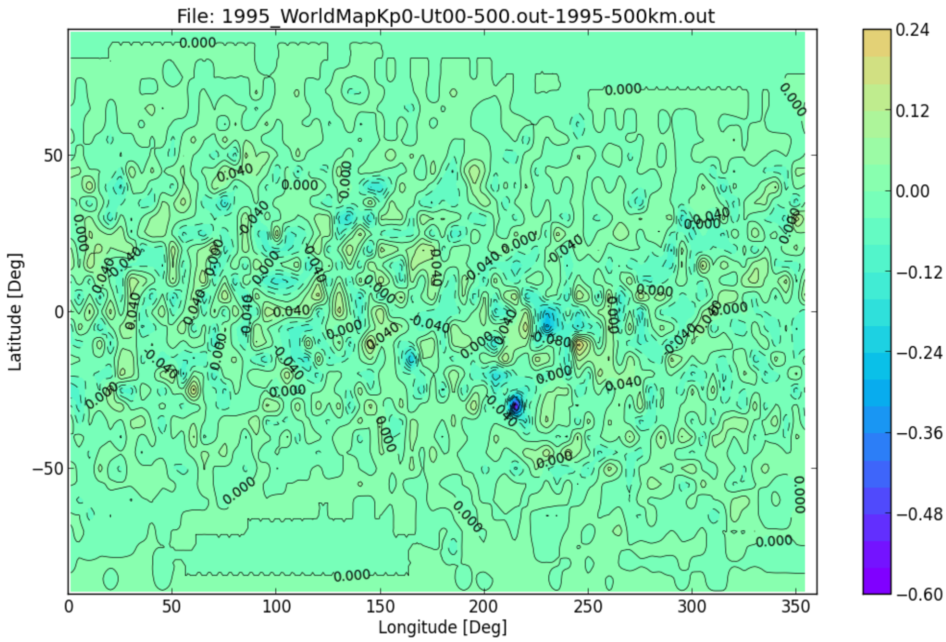

Figure 8: The difference between the MSM and MAGNETOCOSMICS vertical cut-off rigidity (in GV) at global level. There are hot and cold spots in the region around (0˚,240˚), which

are mostly due to poor statistics in the pre-calculated maps and in the MAGNETOCOSMICS map.

The difference in the MSM and MAGNETOCOSMICS calculated vertical cut-off rigidity (in GV) are plotted in Figure 13. One can see There are some hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps and in the MAGNETOCOSMICS map. The larger structure patterns are the real difference between the two predictions. In Figure 14 we plotted the distribution of the relative difference for all positions with Rc > 0.5 GV. Assuming they follow the Gaussian distribution, the corresponding mean is 0.0002 and the standard deviation is 0.015

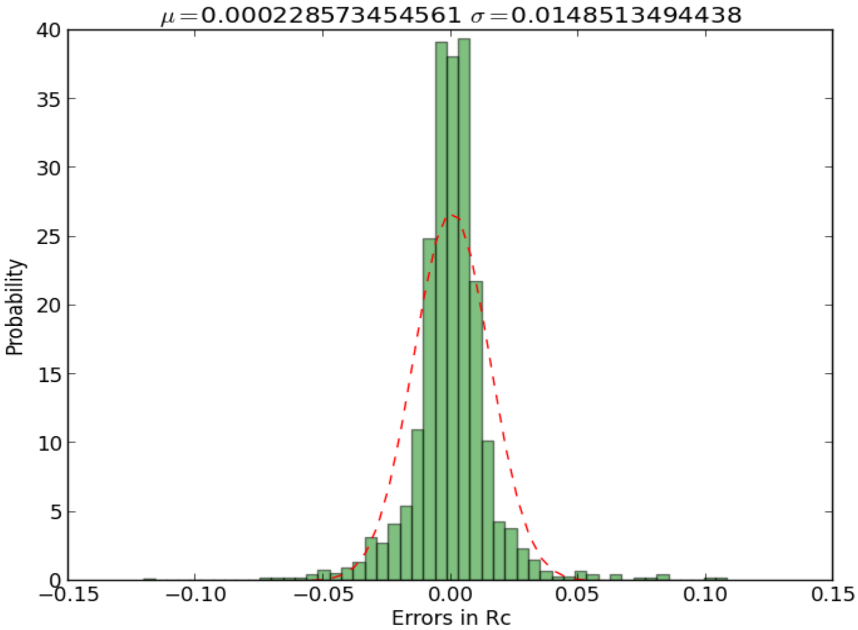

Figure 9: The probability distribution of the differences between the MSM and MAGNETOCOSMICS calculated vertical cut-off rigidity, for locations with Rc > 0.5 GV.

5.1.4 1995.0, UT = 0, Kp = 0, Alt = 1000km

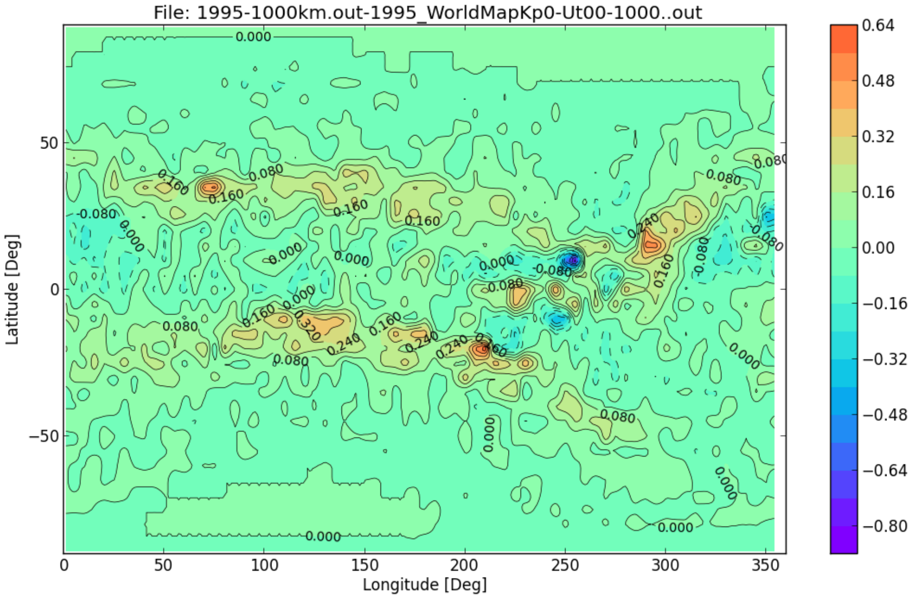

Figure 10: The difference between the MSM and MAGNETOCOSMICS vertical cut-off rigidity (in GV) at global level. There are hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps and in the MAGNETOCOSMICS map.

The difference in the MSM and MAGNETOCOSMICS calculated vertical cut-off rigidity (in GV) are plotted in Figure 15. One can see There are some hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps and in the MAGNETOCOSMICS map. The larger structure patterns are the real difference between the two predictions. In Figure 16 we plotted the distribution of the relative difference for all positions with Rc > 0.5 GV. Assuming they follow the Gaussian distribution, the corresponding mean is 0.001 and the standard deviation is 0.020

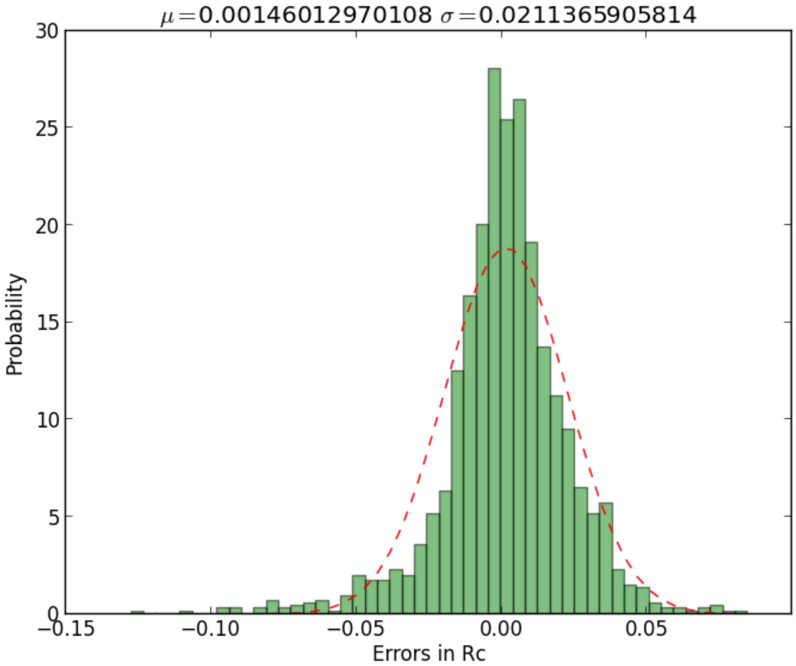

Figure 11: The probability distribution of the differences between the MSM and MAGNETOCOSMICS calculated vertical cut-off rigidity, for locations with Rc > 0.5 GV.

5.1.5 1995.0, UT = 0, Kp = 0, Alt = 5000km

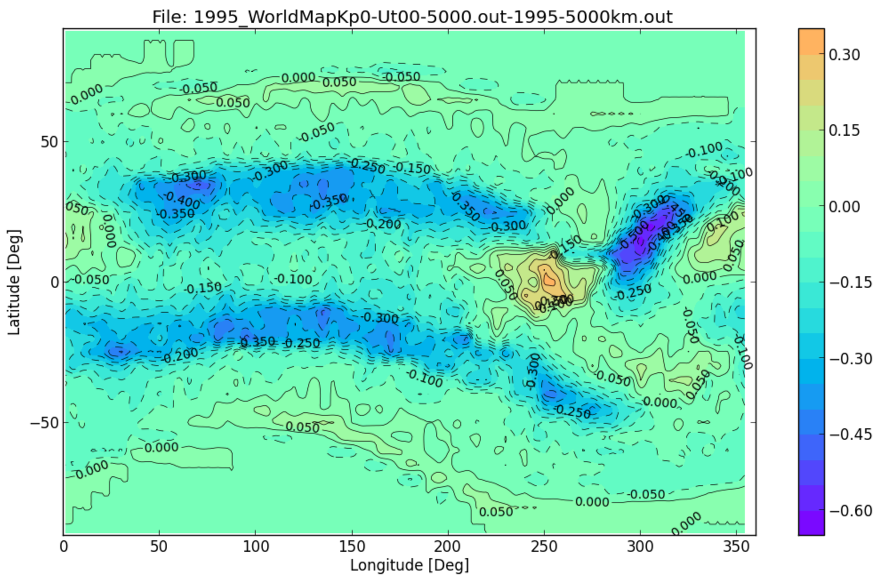

Figure 12: The difference between the MSM and MAGNETOCOSMICS vertical cut-off rigidity (in GV) at global level. There are hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps and in the MAGNETOCOSMICS map.

The difference in the MSM and MAGNETOCOSMICS calculated vertical cut-off rigidity (in GV) are plotted in Figure 17. One can see There are some hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps and in the MAGNETOCOSMICS map. The larger structure patterns are the real difference between the two predictions. In Figure 18 we plotted the distribution of the relative difference for all positions with Rc > 0.1 GV. Assuming they follow the Gaussian distribution, the corresponding mean is 0.05 and the standard deviation is 0.05

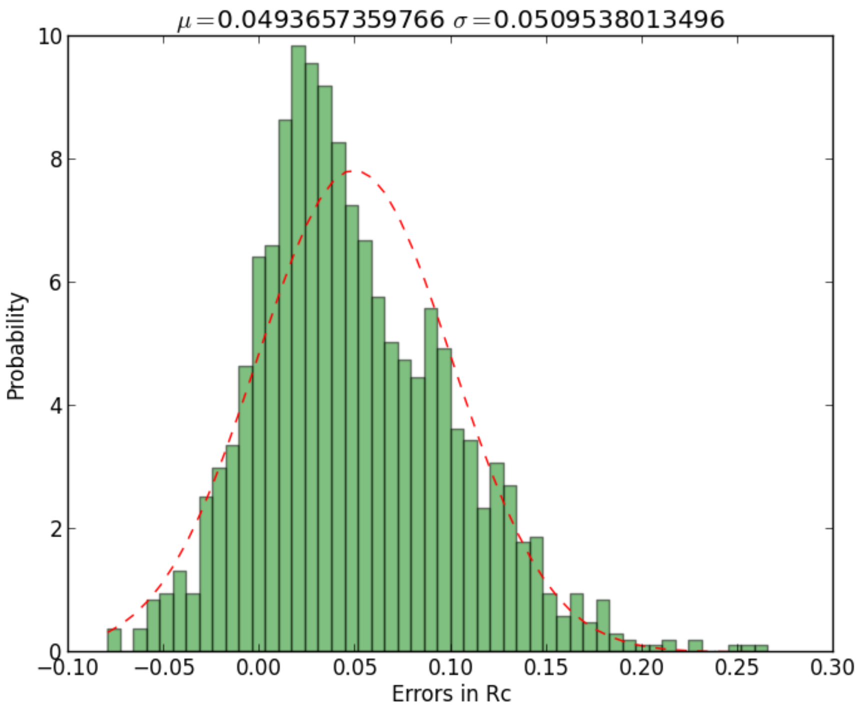

Figure 13: The probability distribution of the differences between the MSM and MAGNETOCOSMICS calculated vertical cut-off rigidity, for locations with Rc > 0.5 GV.

5.1.6 1995.0, UT = 0, Kp = 0, Alt = 10000km

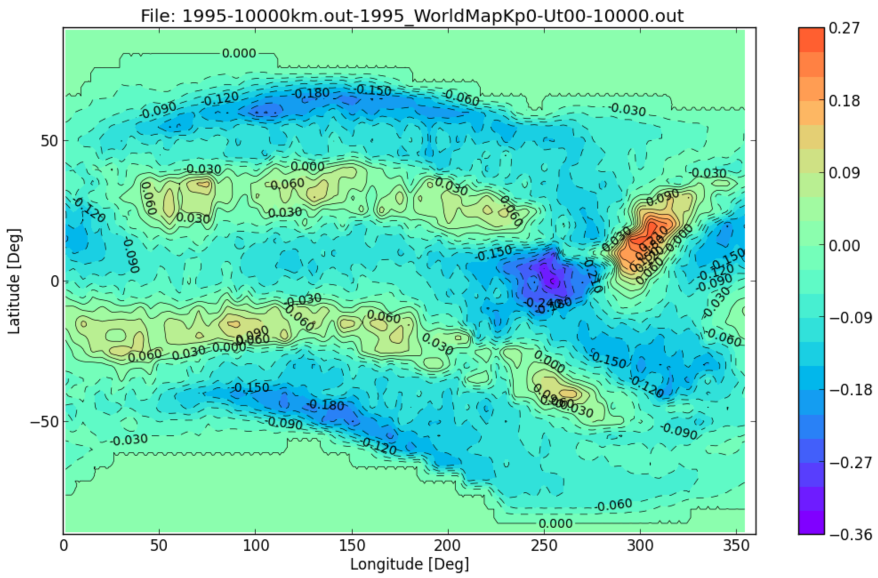

Figure 14: The difference between the MSM and MAGNETOCOSMICS vertical cut-off rigidity (in GV) at global level. There are hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps and in the MAGNETOCOSMICS map.

The difference in the MSM and MAGNETOCOSMICS calculated vertical cut-off rigidity (in GV) are plotted in Figure 19. One can see There are some hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps and in the MAGNETOCOSMICS map. The larger structure patterns are the real difference between the two predictions. In Figure 20 we plotted the distribution of the relative difference for all positions with Rc > 0.1 GV. Assuming they follow the Gaussian distribution, the corresponding mean is -0.15 and the standard deviation is 0.25

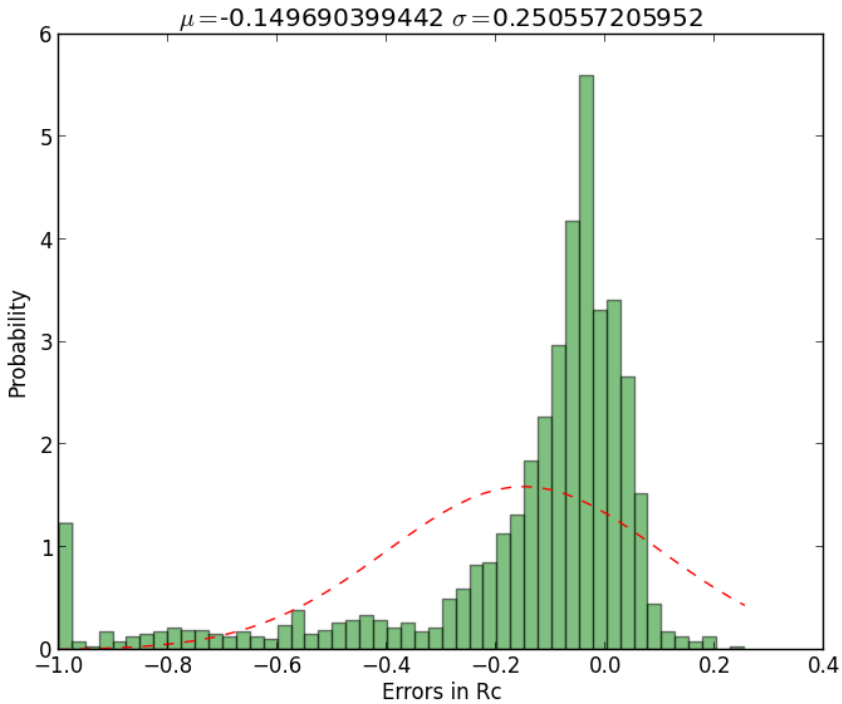

Figure 15: The probability distribution of the differences between the MSM and MAGNETOCOSMICS calculated vertical cut-off rigidity, for locations with Rc > 0.1 GV.

5.2 Against the RCINTUT3 code

In this section we repeat the same comparisons of section 7.1 but with the results produced by the RCINTUT3 code. The table below summaries the derived mean and standard deviation of the distribution:

Altitude (km)	Mean	Standard deviation
20	0.01	0.045
100	0.01	0.043
500	0.01	0.038
1000	0.009	0.033

At 5000 km and above altitudes, the RCINTUT3 code predicted 0 GV cut-off globally, which looks like in correct in the light of recent GOES data.

5.2.1 1995.0, UT = 0, Kp = 0, Alt = 20km

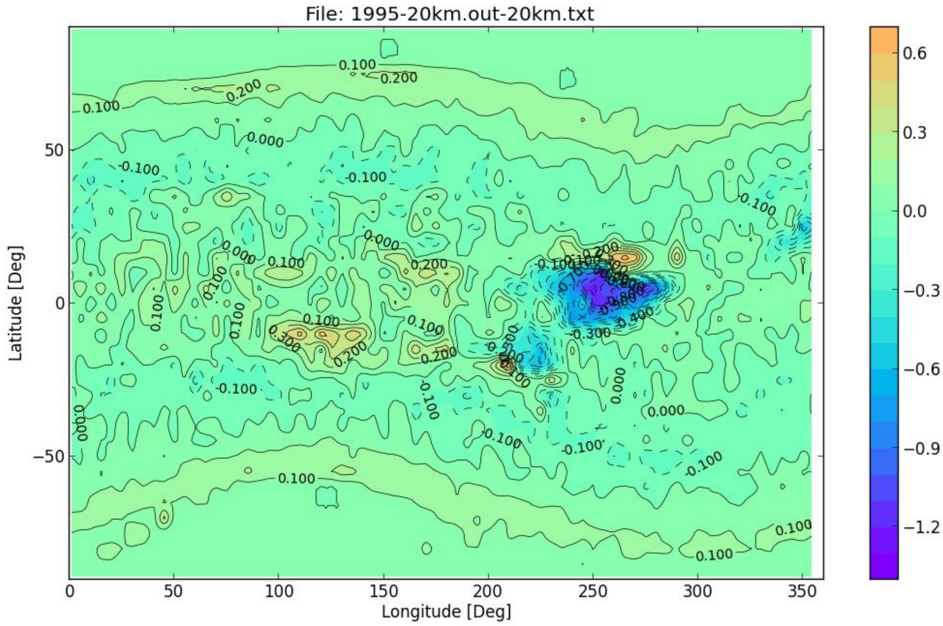

Figure 16: The difference between the MSM and RCINTUT3 vertical cut-off rigidity (in GV) at global level. There are hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps.

The difference in the MSM and RCINTUT3 calculated vertical cut-off rigidity (in GV) are plotted in Figure 21. One can see There are some hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps and also that in the RCINTUT3 maps. The larger structure patterns are the real difference between the two predictions. In Figure 22 we plotted the distribution of the relative difference for all positions with Rc > 0.5 GV. Assuming they follow the Gaussian distribution, the corresponding mean is 0.01 and the standard deviation is 0.044

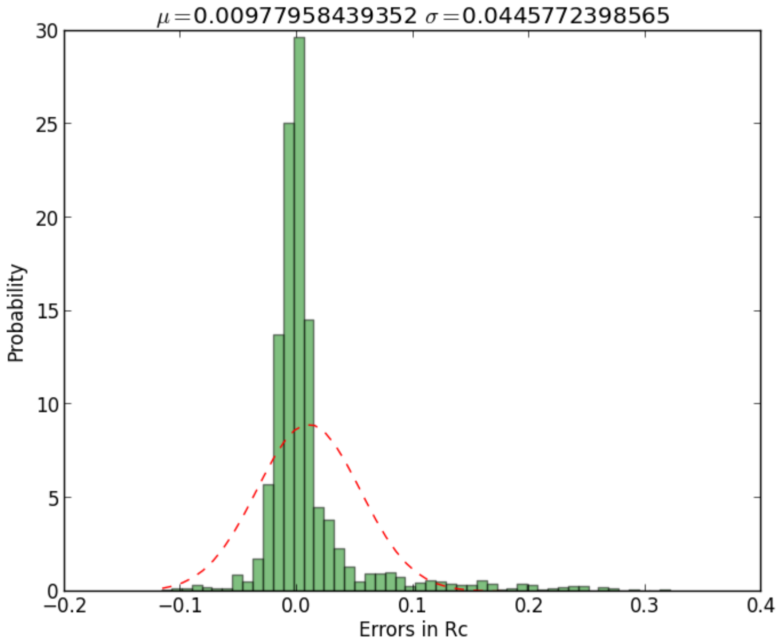

Figure 17: The probability distribution of the differences between the MSM and RCINTUT3 calculated vertical cut-off rigidity, for locations with Rc > 0.5 GV.

5.2.2 1995.0, UT = 0, Kp = 0, Alt = 100km

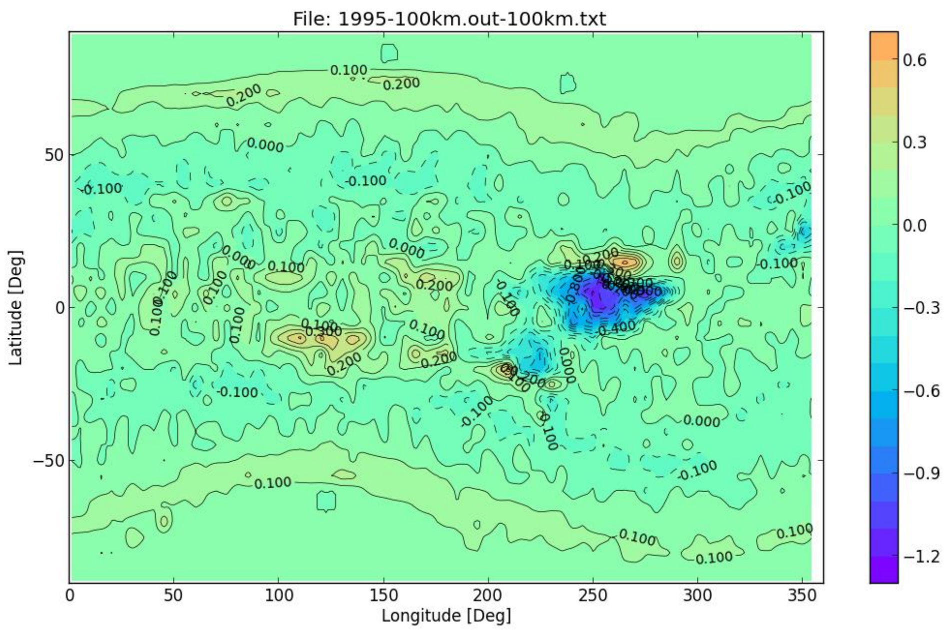

Figure 18: The difference between the MSM and RCINTUT3 vertical cut-off rigidity (in GV) at global level. There are hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps.

The difference in the MSM and RCINTUT3 calculated vertical cut-off rigidity (in GV) are plotted in Figure 23. One can see There are some hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps and that in the RCINTUT3 maps. The larger structure patterns are the real difference between the two predictions. In Figure 24 we plotted the distribution of the relative difference for all positions with Rc > 0.5 GV. Assuming they follow the Gaussian distribution, the corresponding mean is 0.009 and the standard deviation is 0.043

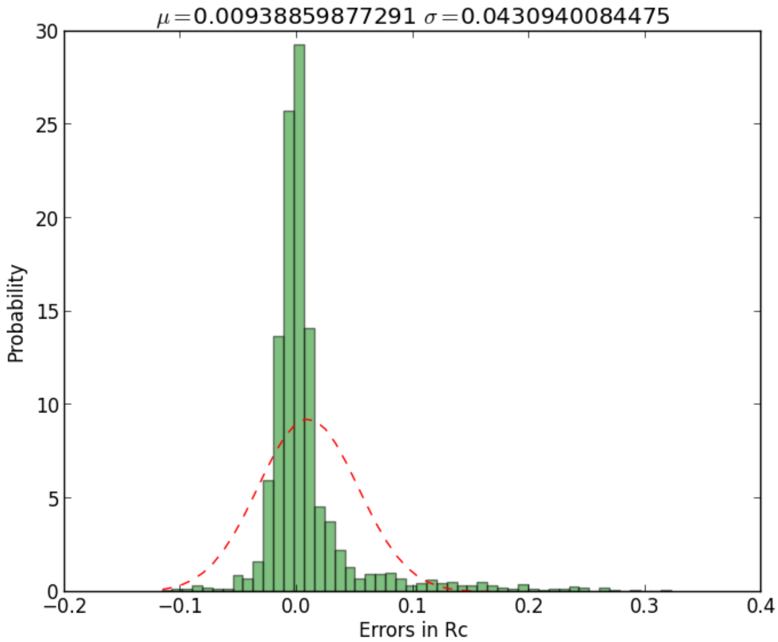

Figure 19: The probability distribution of the differences between the MSM and RCINTUT3 calculated vertical cut-off rigidity, for locations with Rc > 0.5 GV.

5.2.3 1995.0, UT = 0, Kp = 0, Alt = 500km

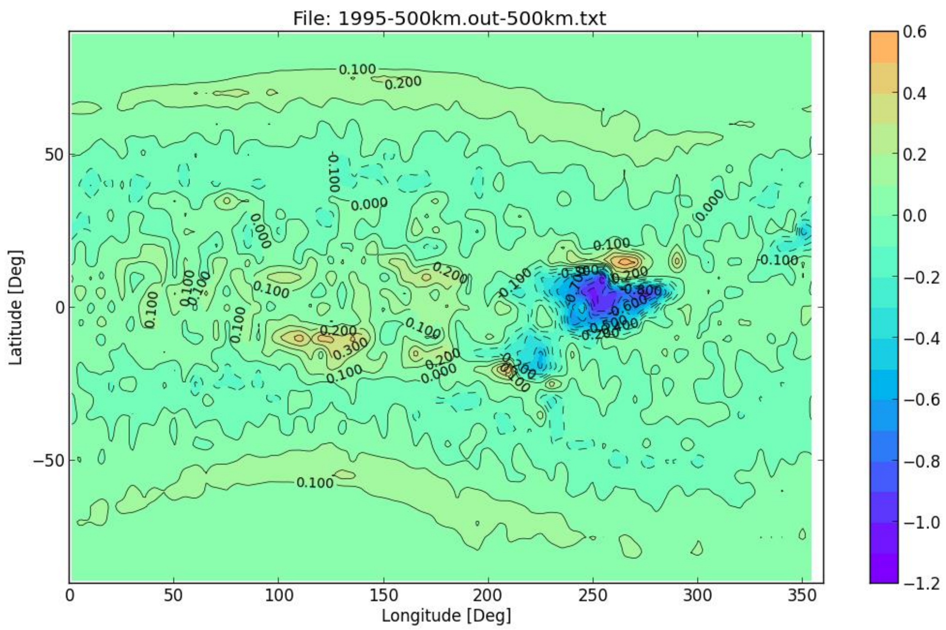

Figure 20: The difference between the MSM and RCINTUT3 vertical cut-off rigidity (in GV) at global level. There are hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps.

The difference in the MSM and RCINTUT3 calculated vertical cut-off rigidity (in GV) are plotted in Figure 25. One can see There are some hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps and that in the RCINTUT3 maps. The larger structure patterns are the real difference between the two predictions. In Figure 26 we plotted the distribution of the relative difference for all positions with Rc > 0.5 GV. Assuming they follow the Gaussian distribution, the corresponding mean is 0.009 and the standard deviation is 0.038

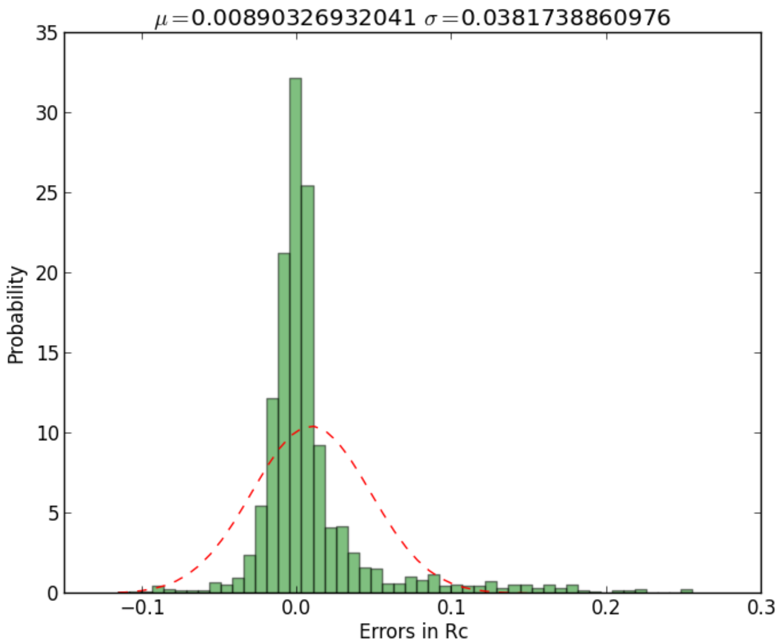

Figure 21: The probability distribution of the differences between the MSM and RCINTUT3 calculated vertical cut-off rigidity, for locations with Rc > 0.5 GV.

5.2.4 1995.0, UT = 0, Kp = 0, Alt = 1000km

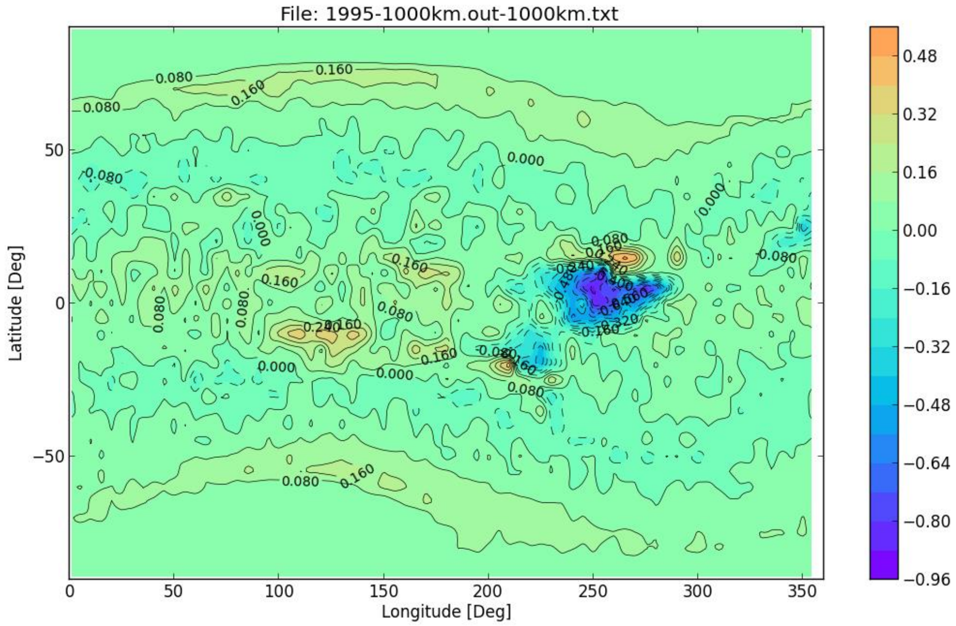

Figure 22: The difference between the MSM and RCINTUT3 vertical cut-off rigidity (in GV) at global level. There are hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps.

The difference in the MSM and RCINTUT3 calculated vertical cut-off rigidity (in GV) are plotted in Figure 27. One can see There are some hot and cold spots in the region around (0˚,240˚), which are mostly due to poor statistics in the pre-calculated maps and that in the RCINTUT3 maps. The larger structure patterns are the real difference between the two predictions. In Figure 28 we plotted the distribution of the relative difference for all positions with Rc > 0.5 GV. Assuming they follow the Gaussian distribution, the corresponding mean is 0.009 and the standard deviation is 0.033

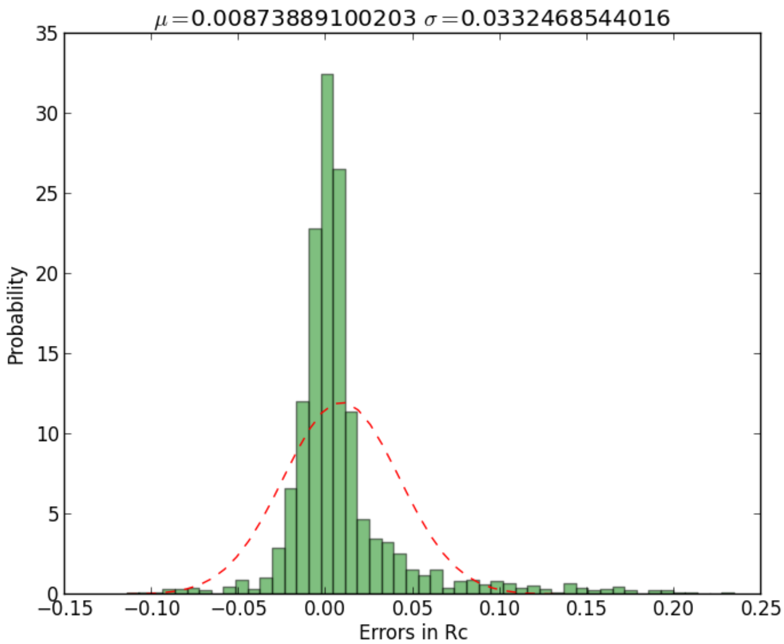

Figure 23: The probability distribution of the differences between the MSM and RCINTUT3 calculated vertical cut-off rigidity, for locations with Rc > 0.5 GV.

5.3 IREM/SREM data

ESA's Standard Radiation Environment Monitor (SREM) has been flown on many ESA missions include Proba1 and INTEGRAL. Proba1 is in a polar orbit (681x561 km) with an inclination angle of 97.9 degrees. INTEGRAL is in a high elliptical orbit with a period of 72 hour. So for a SEP event the SREM on INTEGRAL is most likely measuring the event outside the influence of the magnetosphere, while at the same time the monitor on Proba1 is observing the same event flux with of modulation of the magnetosphere along its orbit. In Figure 29 is an example of the Proba1 and INTEGRAL SREM data of the SEP event on 17th January 2005.The counting rates in three channels are plotted. The INTEGRAL data shows the uninterrupted measurement of the event, while the Prob1 data were modulated by the magnetic shielding. For the time periods when the Proba1 spacecraft is in the polar region the two measurements agree very well.

The Kp index during the day of 17/01/2005 are listed in the table below (data from noaa.gov):

UT (hr)	0	3	6	9	12	15	18	21
Mid-lati	3	2	2	5	5	5	4	3
High-lat	4	4	4	8	8	8	6	4

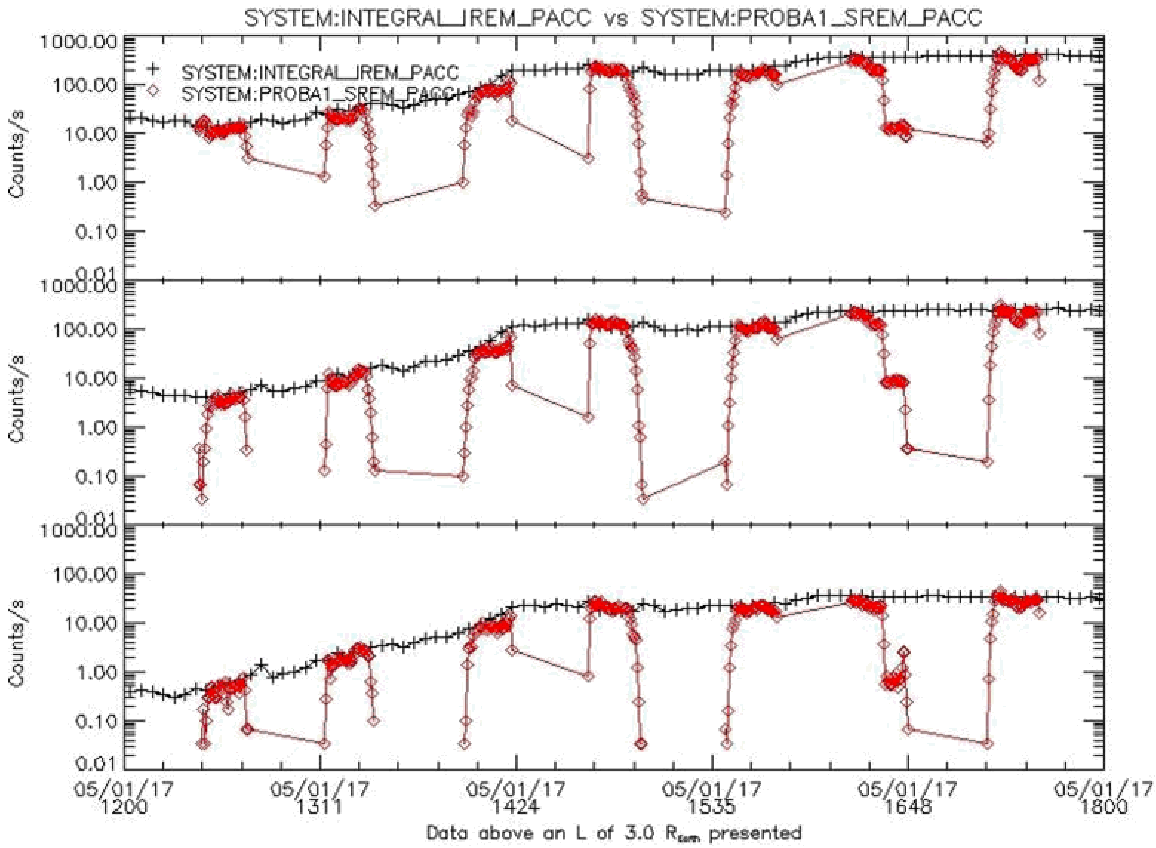

Figure 24: The Proba1 and INTEGRAL SREM data of the SEP event on 17th January 2005.The counting rates in three channels are plotted. The INTEGRAL data shows the uninterrupted measurement of the event, while the Proba1 data were modulated by the magnetic shielding. For the time periods when the Proba1 spacecraft is in the polar regions the two measurements agree very well.

The SREM data channels we selected for this analysis are S34, S13 and C2L, which measure the protons with energies greater than around 12 MeV, 24 MeV and 52 MeV respectively. In Figure 30 are these three data channels between 12:00 and 14:30 hrs. We calculated the transmission functions using the Proba1 orbit data at these energies and they are plotted in Figure 31. Kp index of 5 is used for the calculations. One immediate difference between the SREM data and the transmission functions is that there is a difference of the transition time, say between the 12 MeV and 52 MeV cases predicted by the MSM, but this has not been observed in the SREM data. The most likely explanation is that the thresholds of the SREM channels are not very sharp and there are significant over-laps between the channels.

In Figures 32-34 are Proba1 SREM vs. INTEGRAL IREM data, in channel S34, S13 and C2L respectively. Blue line is the SREM data and red line is the IREM data multiplied by the MSM calculated transmission function with the corresponding Proba1 orbital data.

The meaningful comparison between these two data are at the transition edges when the Probal1 spacecraft entering or leaving the pole regions. The differences during the times over the poles are in fact not related to MSM but statistic or simply due to the in-homogeneity of the event itself at the locations of INTEGRAL and Proba1. It appears the IREM/MSM pole transverse are narrower than the SREM at low energies (S34 and S13) and reasonable agreement is achieved for the C2L (>52 MeV) channel.

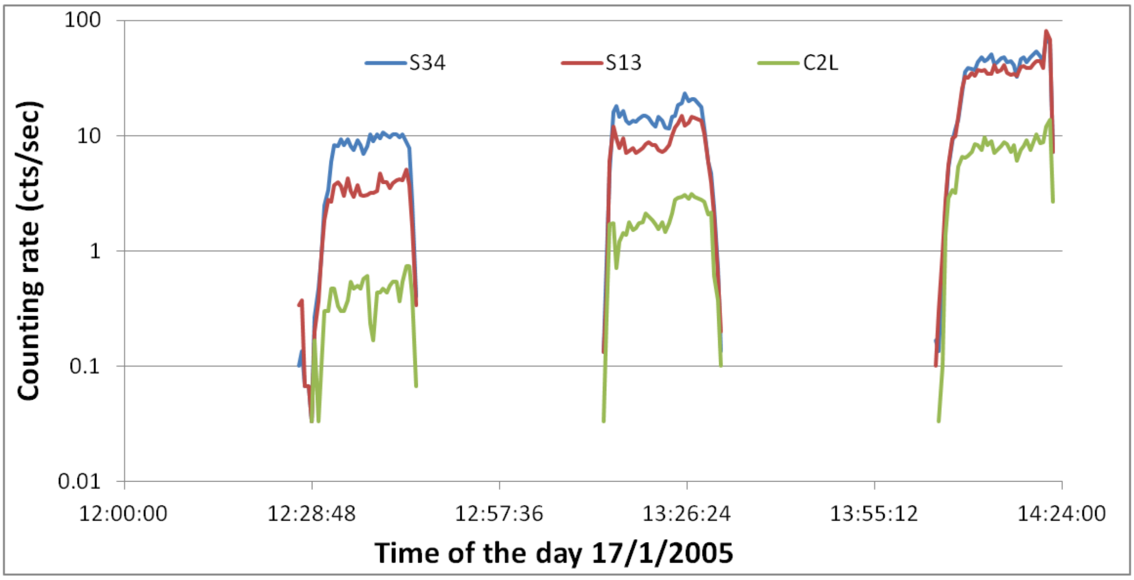

Figure 25: Proba1 SREM data in channel S34, S13 and C2L of the SEP event in 17/01/2005.

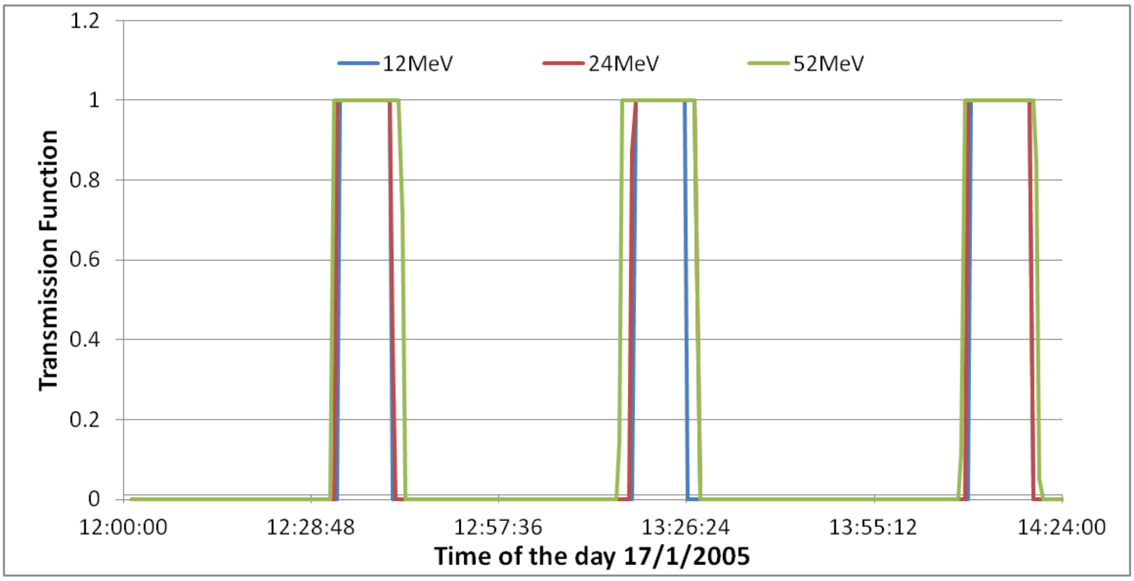

Figure 26: The MSM calculated transmission function for channels S34, S13 and C2L.

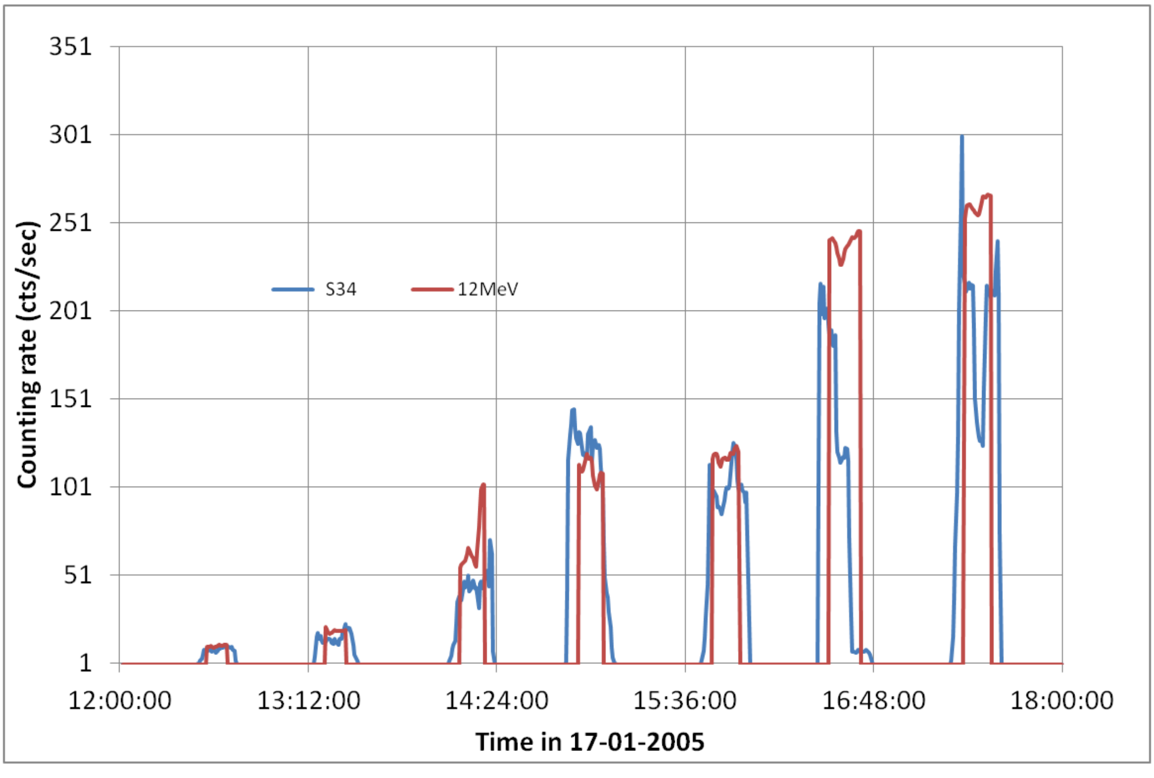

Figure 27: Proba1 SREM vs. INTEGRAL IREM data, in channel S34. Blue line is the SREM data and red line is the IREM data multiplied by the MSM calculated transmission function with the corresponding Proba1 orbital data.

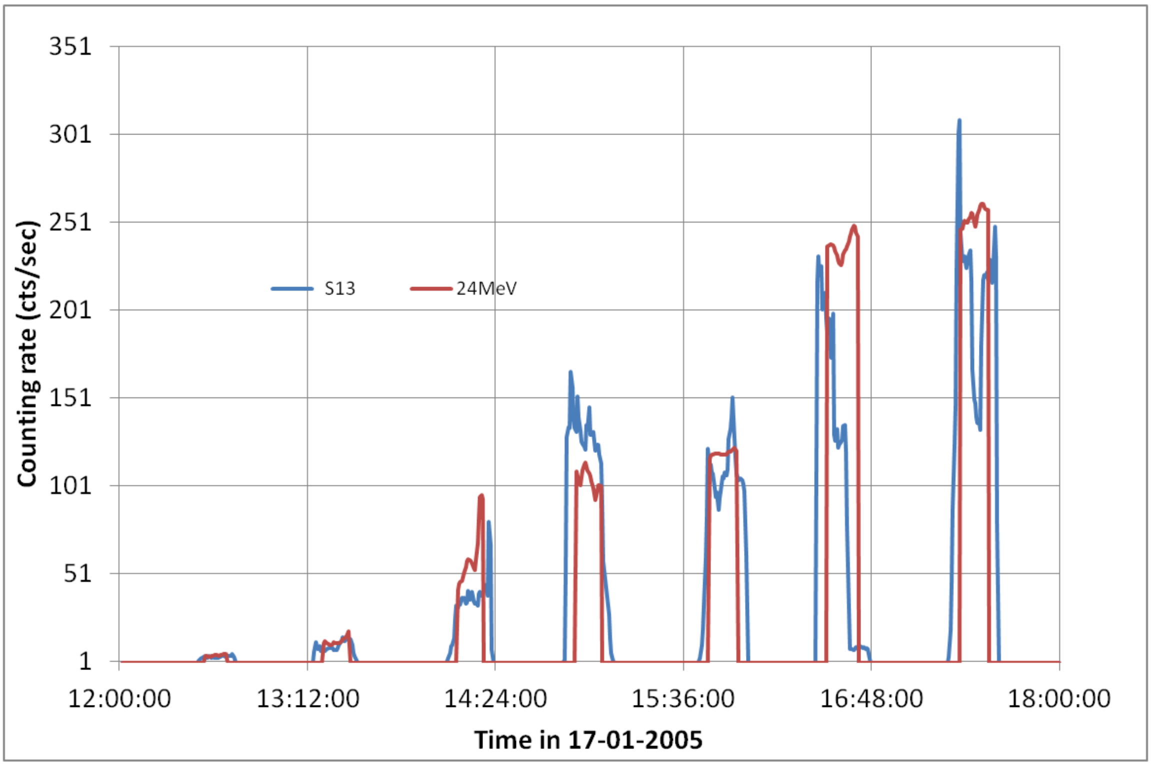

Figure 28: Proba1 SREM vs. INTEGRAL IREM data, in channel S13. Blue line is the SREM data and red line is the IREM data multiplied by the MSM calculated transmission function with the corresponding Proba1 orbital data.

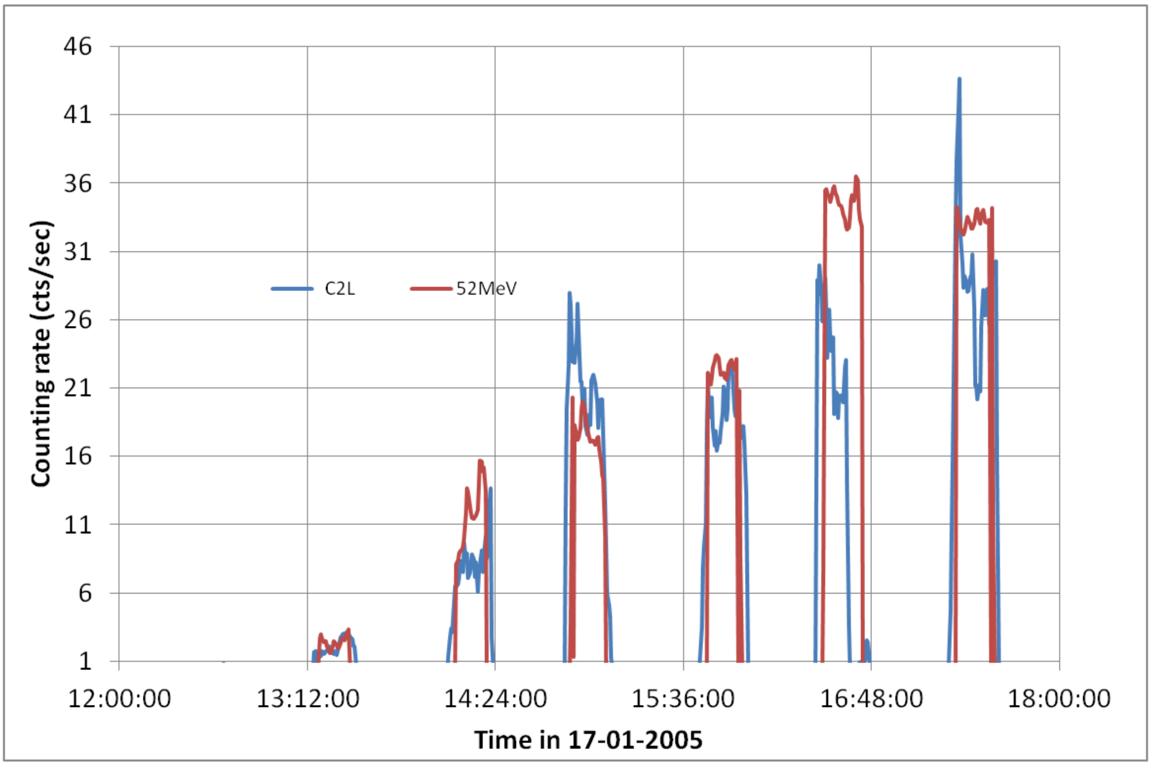

Figure 29: Proba1 SREM vs. INTEGRAL IREM data, in channel C2L. Blue line is the SREM data and red line is the IREM data multiplied by the MSM calculated transmission function with the corresponding Proba1 orbital data.

5.4 SAMPEX data

SAMPEX is in a low earth orbit at an altitude of 520 by 670 km and 82 degrees inclination. We selected the PET proton data for the date 17/01/2005. The data are available in differential proton flux ($cm^{-2}\ s^{-1}\ sr^{-1}\ MeV^{-1}$) in 15 channels, in every 6 seconds.

The energy bins are: 19.0- 21.1 MeV, 21.1- 23.2 MeV, 23.2- 25.2 MeV, 25.2- 27.4 MeV, 27.4- 37.4 MeV, 37.4- 45.8 MeV, 45.8- 53.0 MeV, 53.0- 65.2 MeV, 65.2- 70.5 MeV, 70.5- 76.1 MeV, 76.1- 85.1 MeV, 85.1-120.0 MeV, 120.0-200.0 MeV, 200.0-300.0 MeV, 300.0- 500.0 MeV.

Figure 35 shows examples of the proton fluxes of the 17/01/2005 SEP event in 3 channels. Please note in addition to the SEP protons measured during the pole passages, there are also SSA trapped proton around the time 13:20, 15:00, 16:40, 18:20. These SAA protons can be filtered out with a simple restriction of the L > 3 Re. Another thing worth noting is that we only plotted the none-zero SAMPEX data points. In fact there are zero data points in the SAMPEX data, in the high energy channels in particular, due to the high time resolution (6 s) of the data.

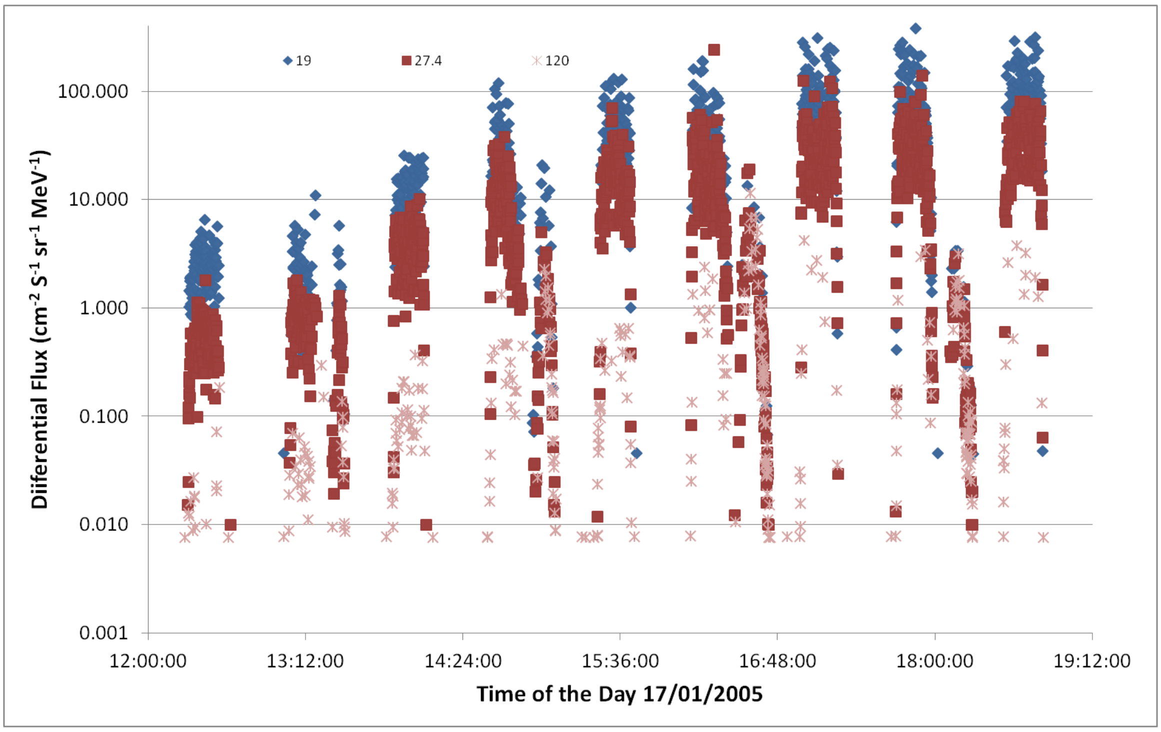

Figure 30: SAMPEX data of the 17/01/2005 SEP event in 3 channels: 19.0- 21.1 MeV, 27.4- 37.4 MeV, and 120.0-200.0 MeV.

As in the case of the SREM data, we calculated the SAMPEX transmission function using MSM with its orbit data. In Figures 36 -49, we plotted the SAMPEX data in an individual channel, plus the MSM calculated transmission function times 1/2 of the max SAMPEX flux during the time period of the plot. Also the filter L > 3 Re is applied to the SAMPEX data. The following can be observed between the MSM transmission functions and SAMPEX data:

1. Similar to the case in Proba1/SREM data, the agreement are reasonable in the high energy channels, i.e. > ~ 70 MeV. Below this energy the MSM transitions are narrower than observed.
2. The passage around the 13:10 hour is problematic for MSM, even in high energy channels. MSM under predicted the transmission in all channels and in particular

no transmission below 30 MeV, while SAMPEX has clear detections. This is could be due to the fact the passage is in the region very close to the SSA (see Figure 37), and the magnetic field/MAGNETOCOSMICS code may have bigger uncertainties in making predictions in this region. This is also reflected in the pre-calculated maps, e.g. see Figure 4.

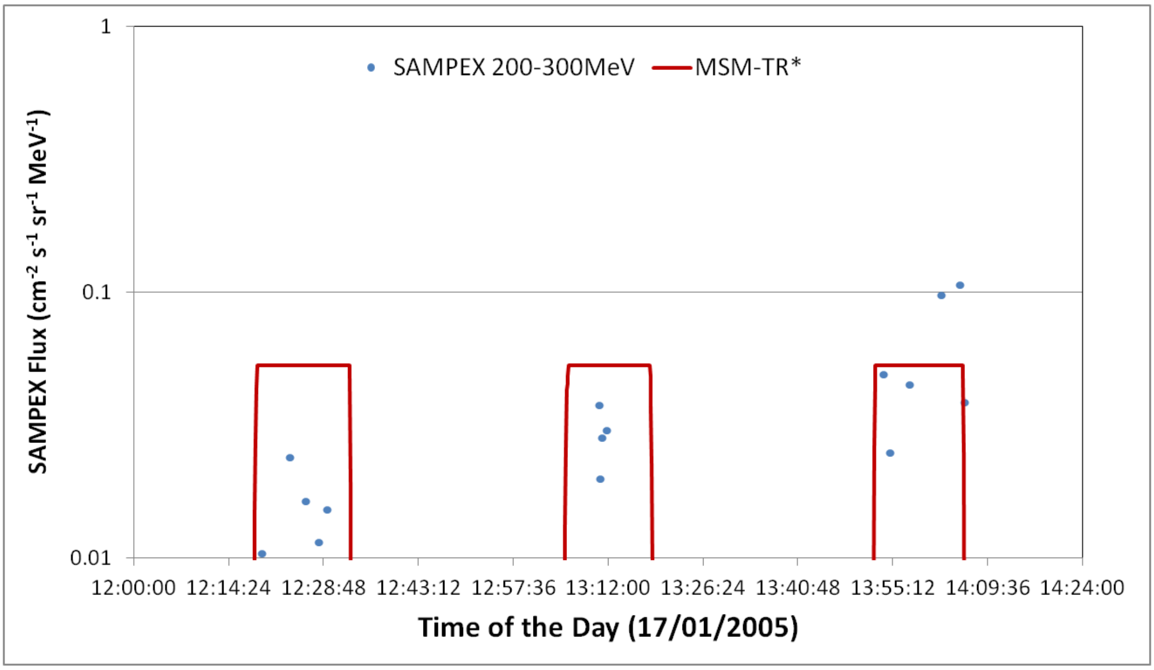

Figure 31: SEP proton fluxes between 200-300 MeV as derived from SAMPEX PET data. Also plotted is the MSM calculated transmission function (times 1/2 of the max of the measured fluxes).

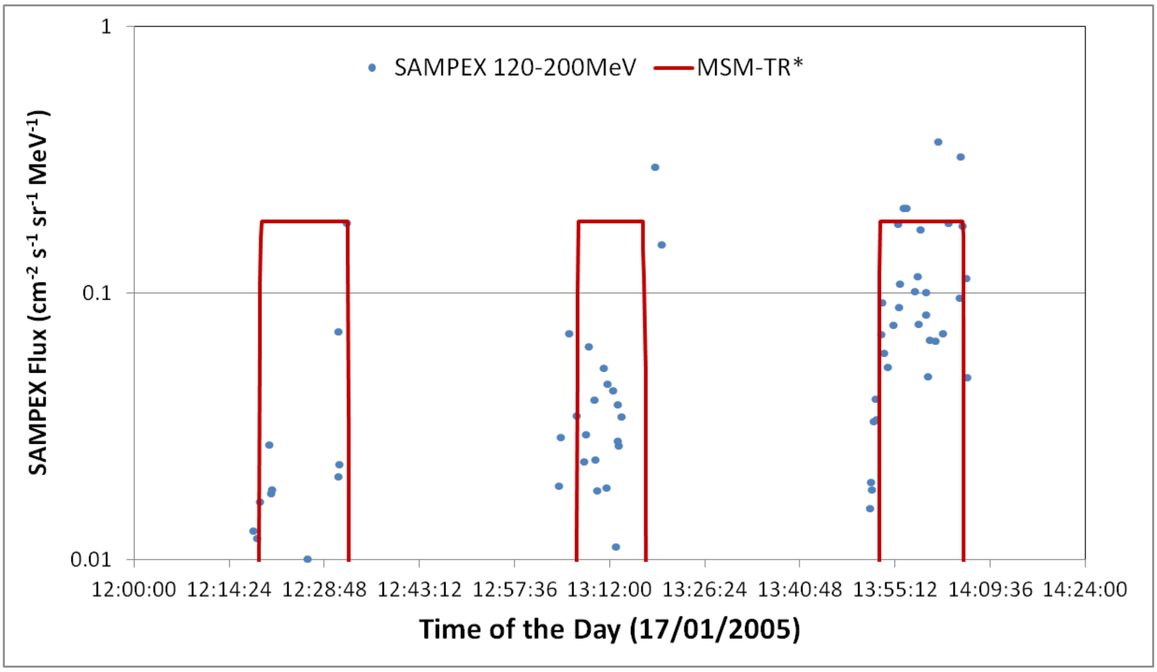

Figure 32: SEP proton fluxes between 120-200 MeV as derived from SAMPEX PET data. Also plotted is the MSM calculated transmission function (times 1/2 of the max of the measured fluxes).

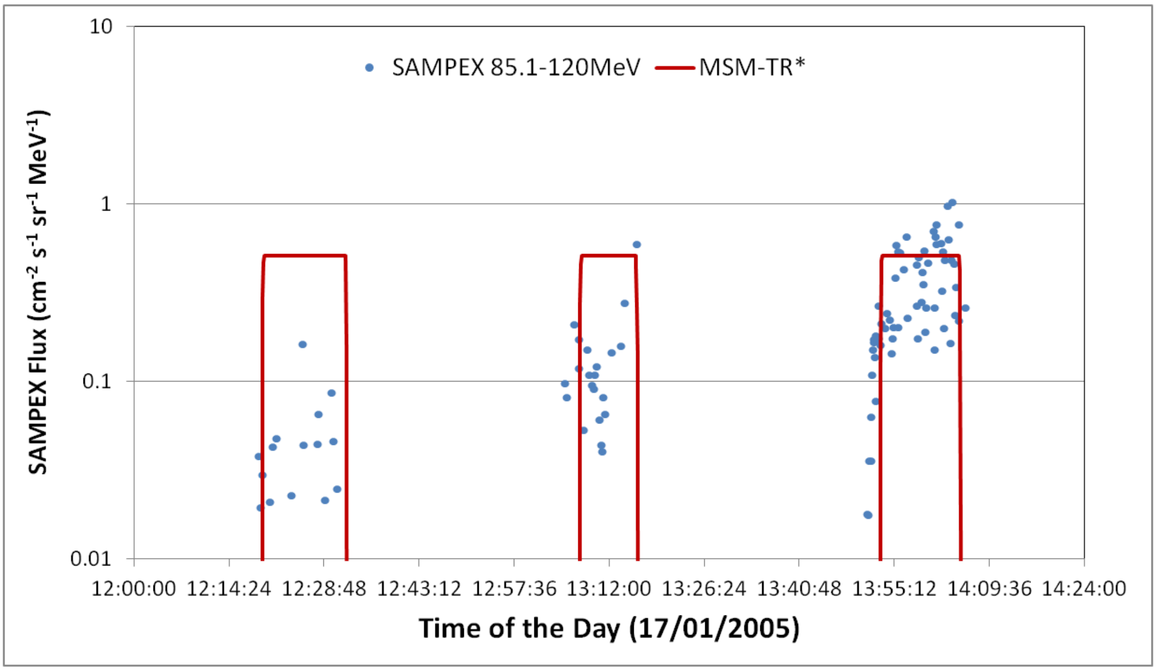

Figure 33: SEP proton fluxes between 85.1-120 MeV as derived from SAMPEX PET data. Also plotted is the MSM calculated transmission function (times 1/2 of the max of the measured fluxes).

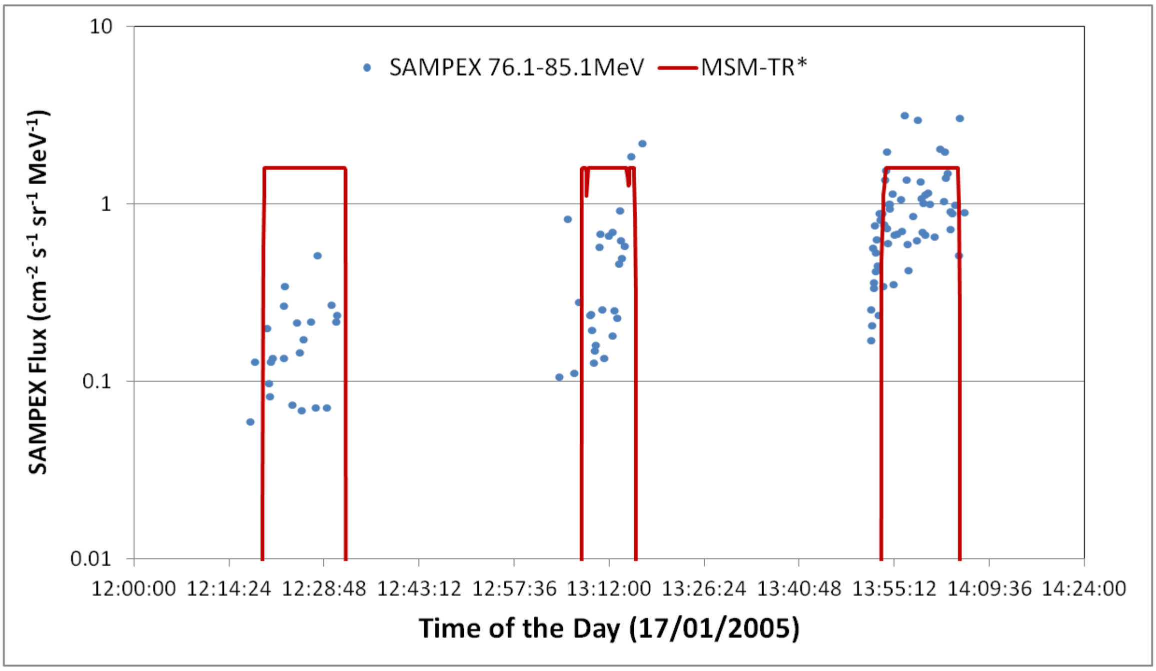

Figure 34: SEP proton fluxes between 76.1-85.1 MeV as derived from SAMPEX PET data. Also plotted is the MSM calculated transmission function (times 1/2 of the max of the measured fluxes).

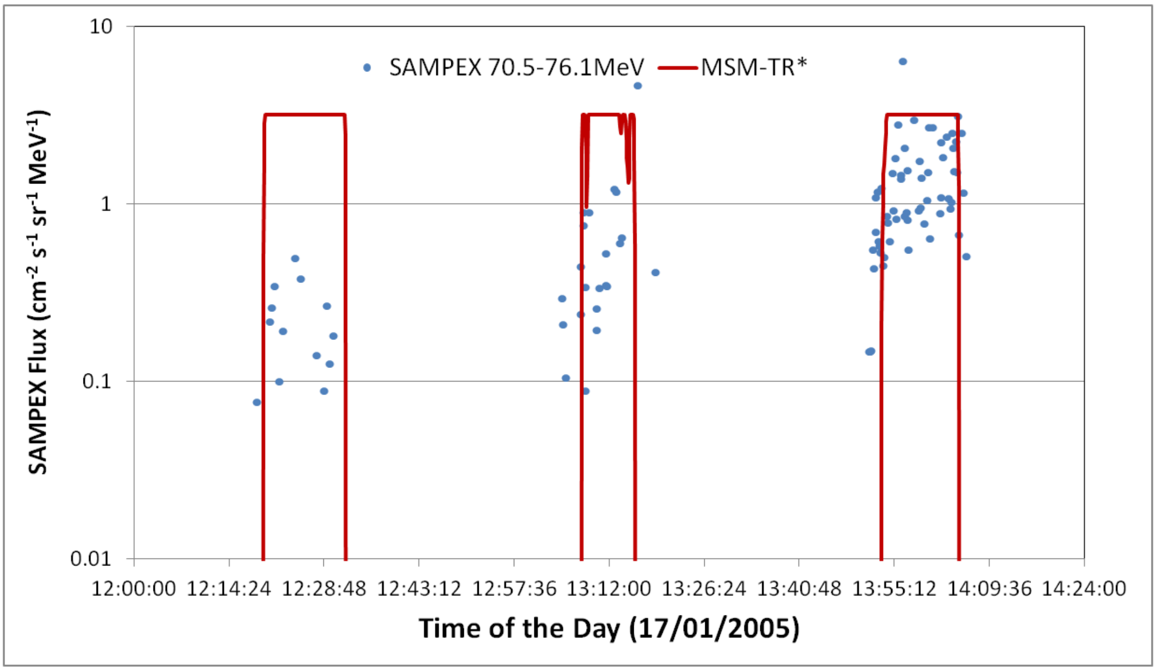

Figure 35: SEP proton fluxes between 70.5-76.1 MeV as derived from SAMPEX PET data. Also plotted is the MSM calculated transmission function (times 1/2 of the max of the measured fluxes).

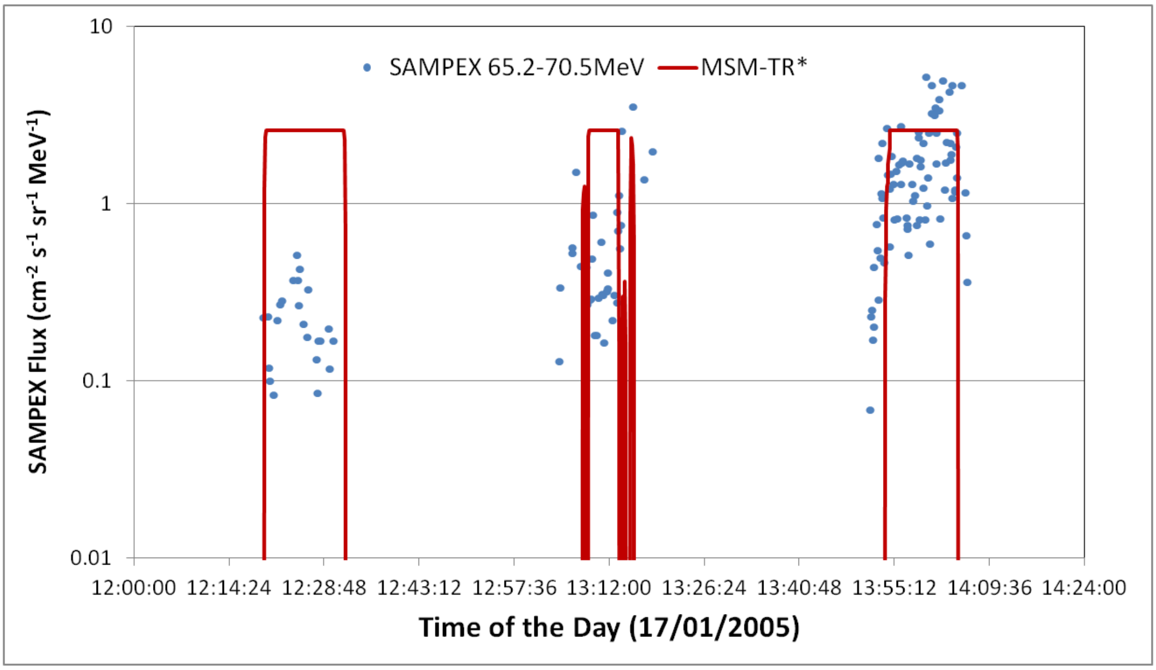

Figure 36: SEP proton fluxes between 65.2-70.5 MeV as derived from SAMPEX PET data. Also plotted is the MSM calculated transmission function (times 1/2 of the max of the measured fluxes).

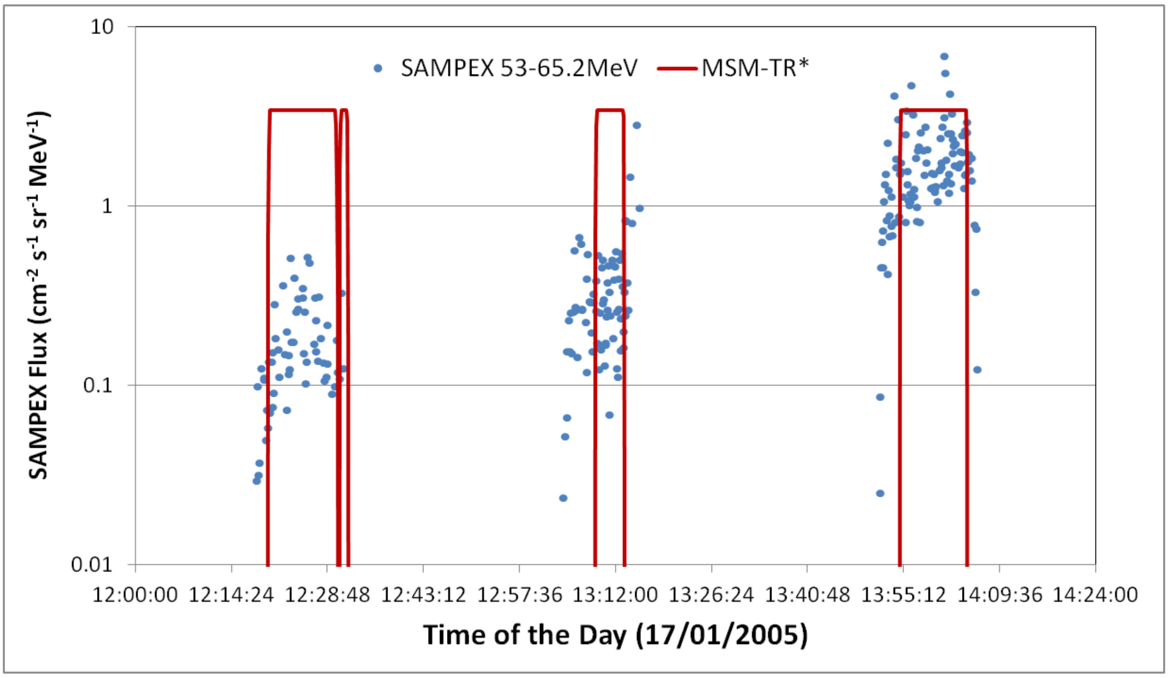

Figure 37: SEP proton fluxes between 53-65.2 MeV as derived from SAMPEX PET data. Also plotted is the MSM calculated transmission function (times 1/2 of the max of the measured fluxes).

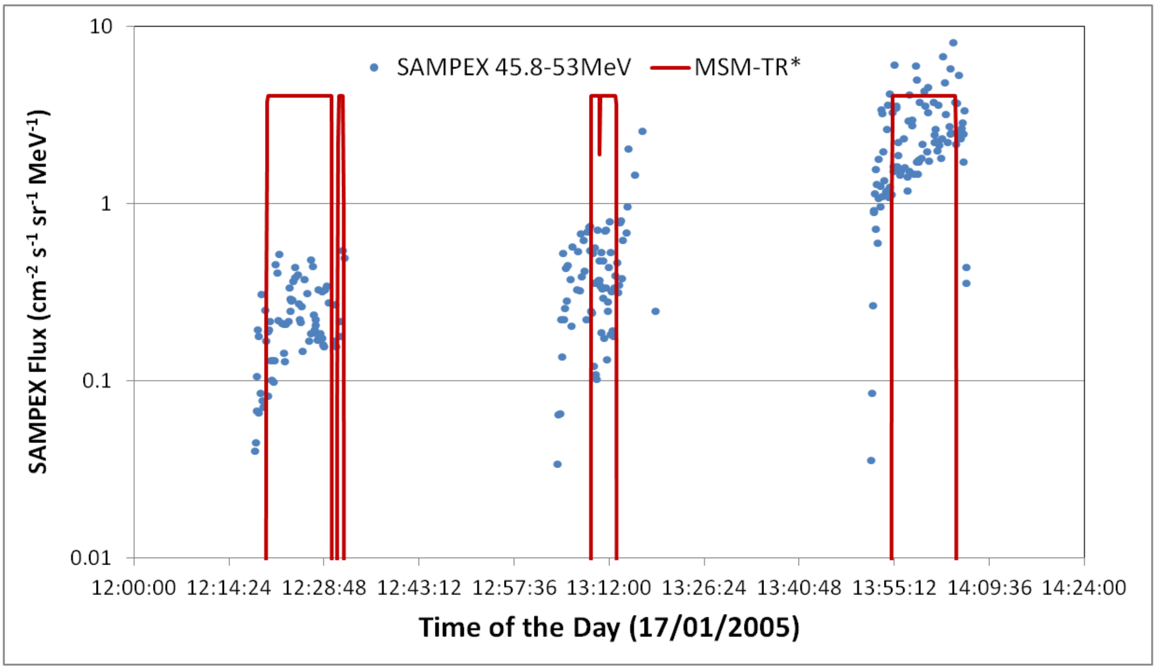

Figure 38: SEP proton fluxes between 45.8-53 MeV as derived from SAMPEX PET data. Also plotted is the MSM calculated transmission function (times 1/2 of the max of the measured fluxes).

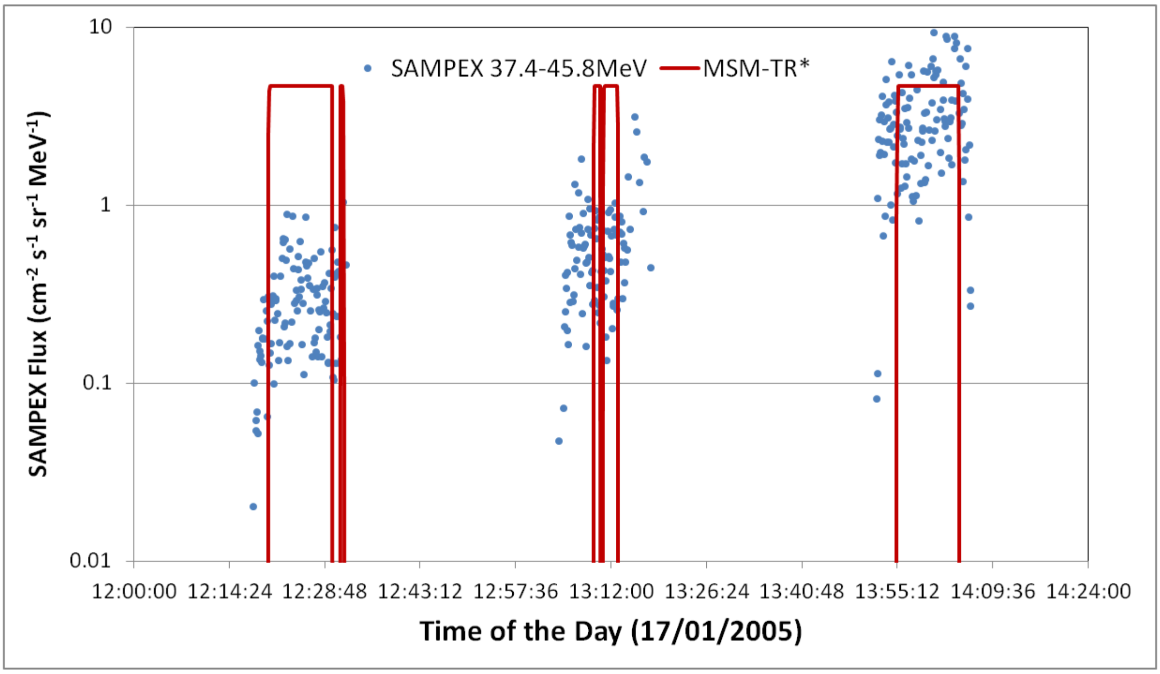

Figure 39: SEP proton fluxes between 37.4-45.8 MeV as derived from SAMPEX PET data. Also plotted is the MSM calculated transmission function (times 1/2 of the max of the measured fluxes).

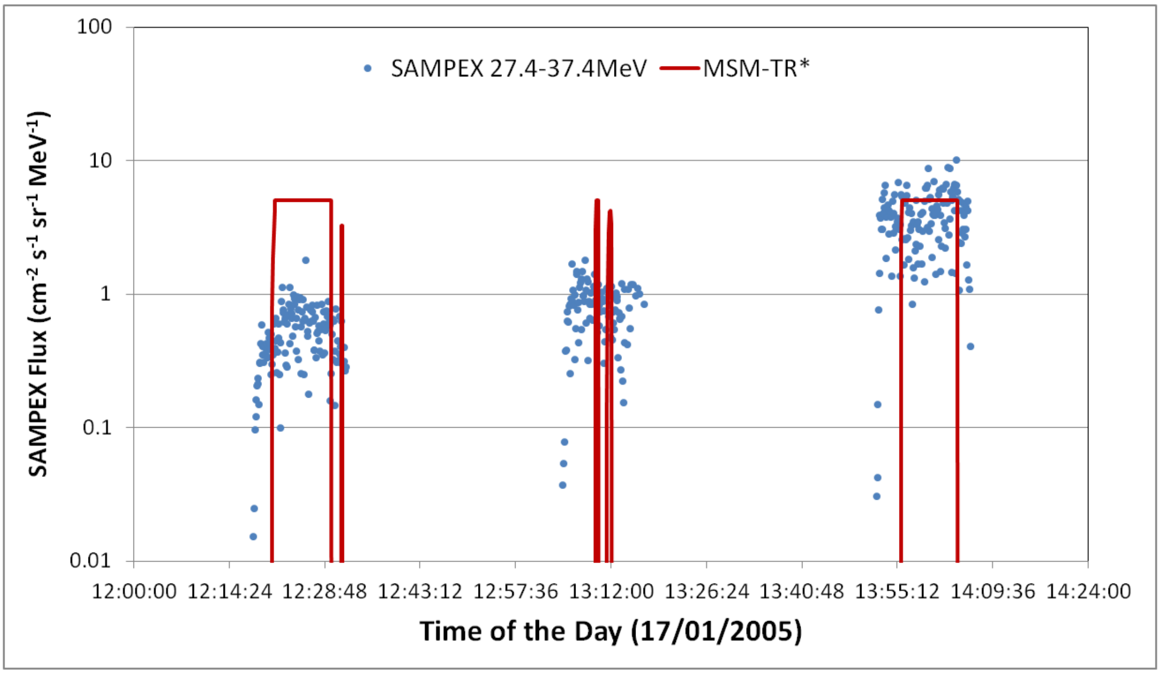

Figure 40: SEP proton fluxes between 27.4-37.4 MeV as derived from SAMPEX PET data. Also plotted is the MSM calculated transmission function (times 1/2 of the max of the measured fluxes).

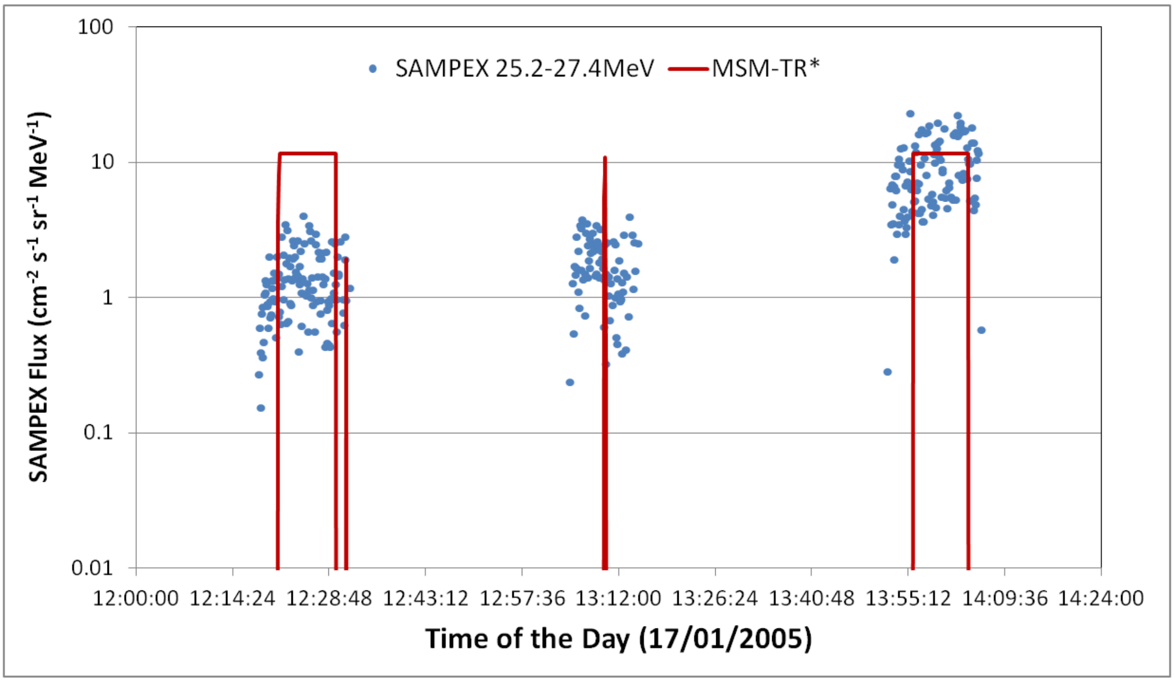

Figure 41: SEP proton fluxes between 25.2-27.4 MeV as derived from SAMPEX PET data. Also plotted is the MSM calculated transmission function (times 1/2 of the max of the measured fluxes).

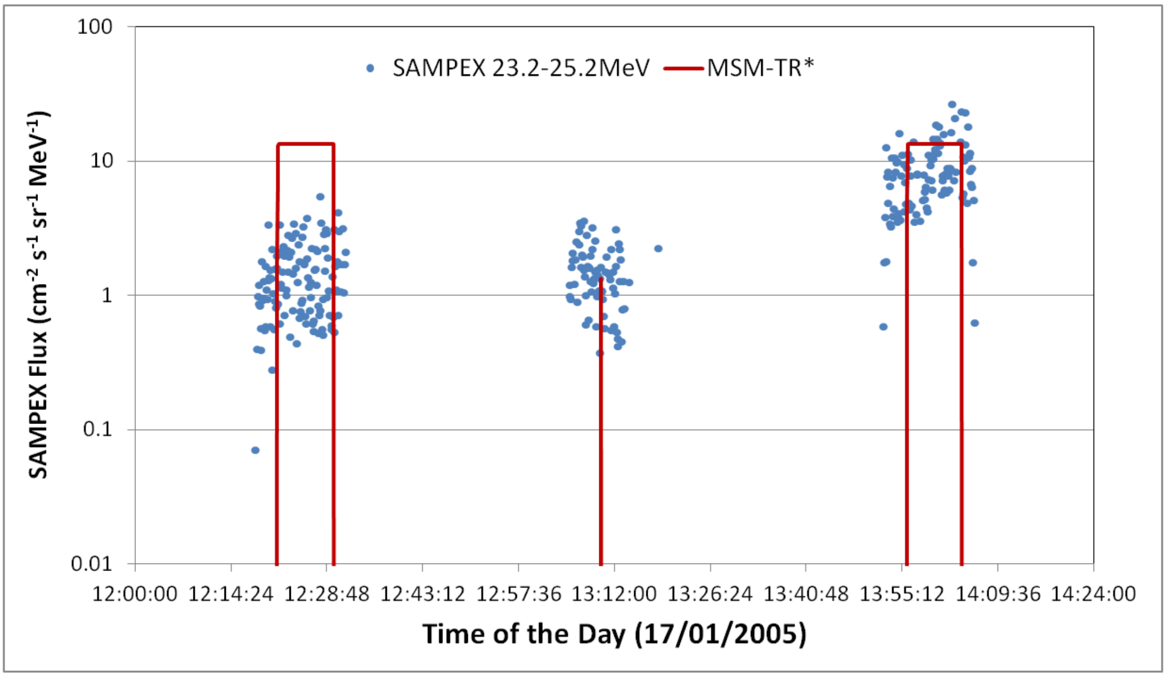

Figure 42: SEP proton fluxes between 23.2-25.2 MeV as derived from SAMPEX PET data. Also plotted is the MSM calculated transmission function (times 1/2 of the max of the measured fluxes).

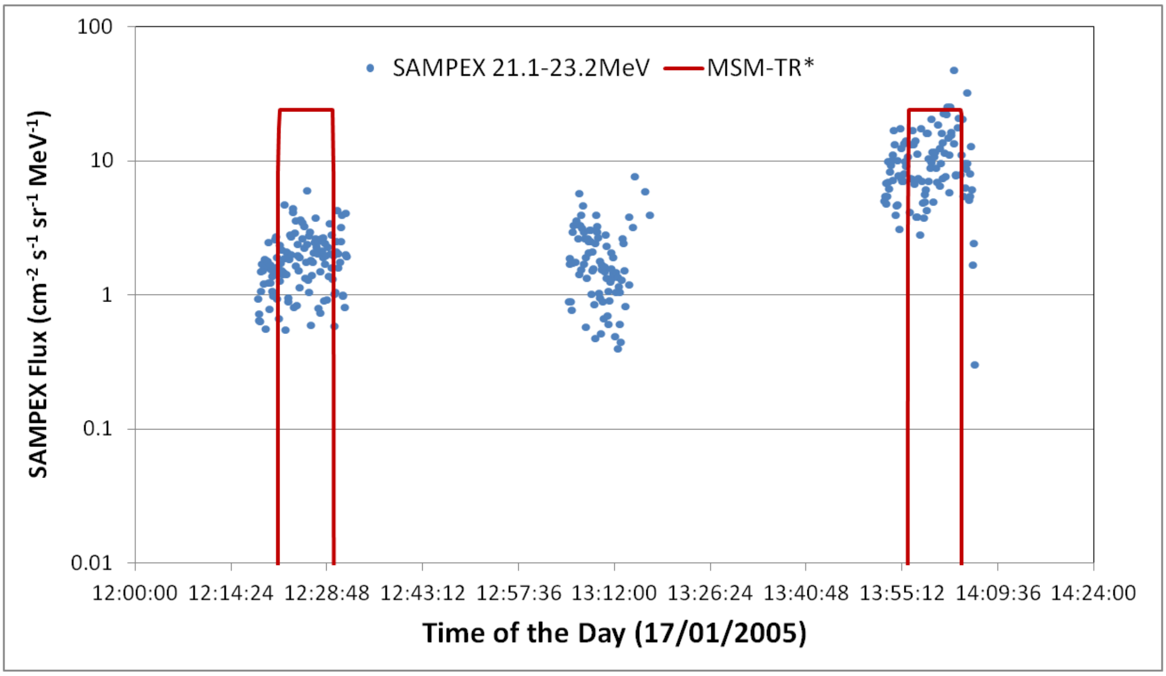

Figure 43: SEP proton fluxes between 21.2-23.2 MeV as derived from SAMPEX PET data. Also plotted is the MSM calculated transmission function (times 1/2 of the max of the measured fluxes).

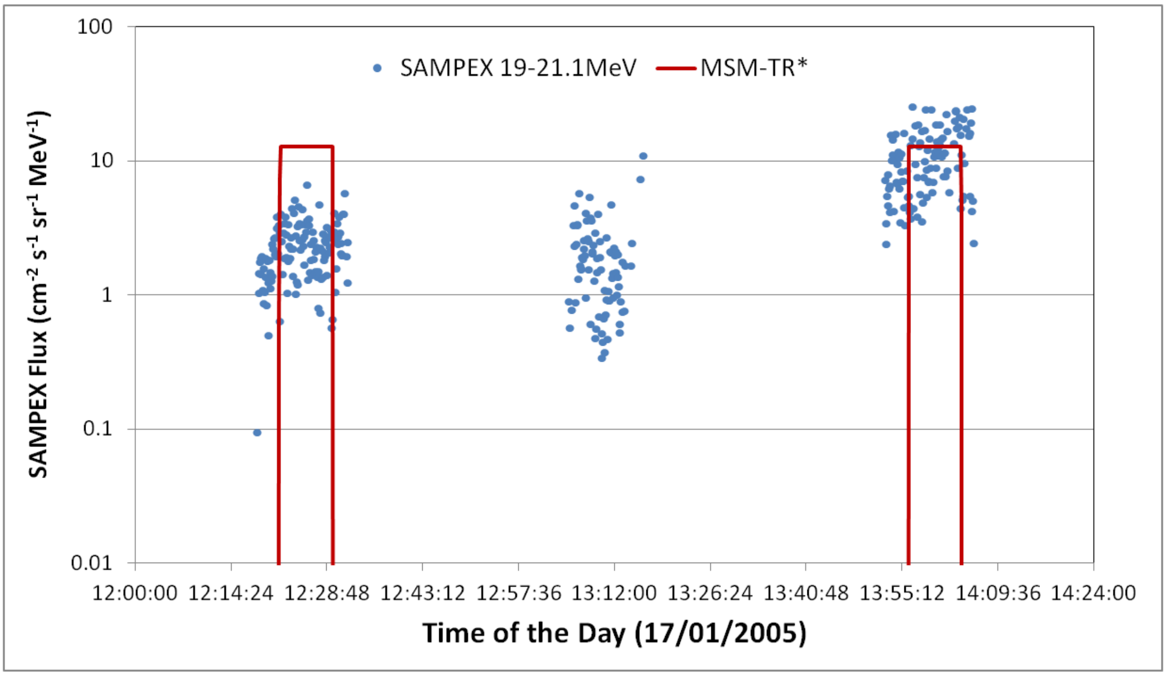

Figure 44: SEP proton fluxes between 19-21.1 MeV as derived from SAMPEX PET data. Also plotted is the MSM calculated transmission function (times 1/2 of the max of the measured fluxes).

5.5 Error Analysis

From section 7.1 we have:

ΔR_{VC}:

- MGNETOCOSMICS intrinsic resolution 0.01 GV,
- typical error level ~ 1%

ΔR_C:

- ~5x the $\Delta R_{VC,}$ i.e.:
- max(5%*Rc, 0.05)

Calculation for each orbital position P:

- MSM generates Ta(R), the averaged transmission function
- ΔR = max(a%*R, b), default a=5, b=0.05
- ΔTa(R) = 0, if Ta(R) = 1, or 0
- ΔTa(R) = (Ta(R+ΔR) – Ta(R-ΔR))/2, otherwise
- ΔTa(R) can be treated as the standard deviation of Ta(R) ? This is nor true!

6 Discussions

6.1 Radial distance adjustment

In RCINTUT3 there is a radial distance adjustment to the algorithm. The reason is explained in the email from Don Smart (09/12/2013): " If you examine the cut-off interpolation FORTRAN code in detail, you may notice that there is an adjustment in the "L" value altitude interpolation process at the end of Subroutine LINT5X5 in a section labeled "Radial Distance Adjustment". While the "L" interpolation equation has the basic form of L**-2, when this exact form is used to extend to geosynchronous altitude, the cut-off values extrapolated from the near Earth low altitudes are too high; approximately 0.3 GV at the magnetic equator. (See figure 7 of Shea & Smart, JGR, 72, 3447, 1967). We incorporated a "patch" (actually an "ad-hoc" exponential function that makes adjustments so at 6.6 earth radii the vertical cut-off rigidity for local noon at the magnetic equator under extremely quiet magnetic conditions is about zero (or extremely small). This "ad-hoc" exponential function is not going to be reliable beyond geosynchronous distances."

The current understanding of the cut-off at geosynchronous altitudes is that they can be non-zero, as indicated by the "East-West" effect observed in the GOES solar proton data. We performed MAGNETOCOSMICS simulations to calculated the detailed cut-off maps at GEO altitude. Figure 45 is such a map for epoch 1995.0, UT at 0 hour and Kp = 0, and Figure 51 is a similar one but for UT=6hr and Kp=1. They show that at the equatorial plane there is can be regions with cut-off up to ~0.4 GV. This however is the vertical cut-off and it implies the cut-off for charged particles coming in from the East direction will be ~ 1.6 GV which is too high! In the GOES SEP data significant "East-West" effect has been observed in solar protons up to ~ 0.4 GV - indicating a vertical cut-off ~0.1 GV only. To take this into account, the "radial correction" as in RCINTUT3 correction has been revised so it reduces the vertical cut-off rigidity at 3600 km down to the ~ 0.1 GV level in the MSM tool, rather than ~ 0 GV. Specifically the correction factor is now calculated as RCORT = (LOG(RADISTSQ))/14.0, instead of RCORT = (LOG(RADISTSQ))/12.0

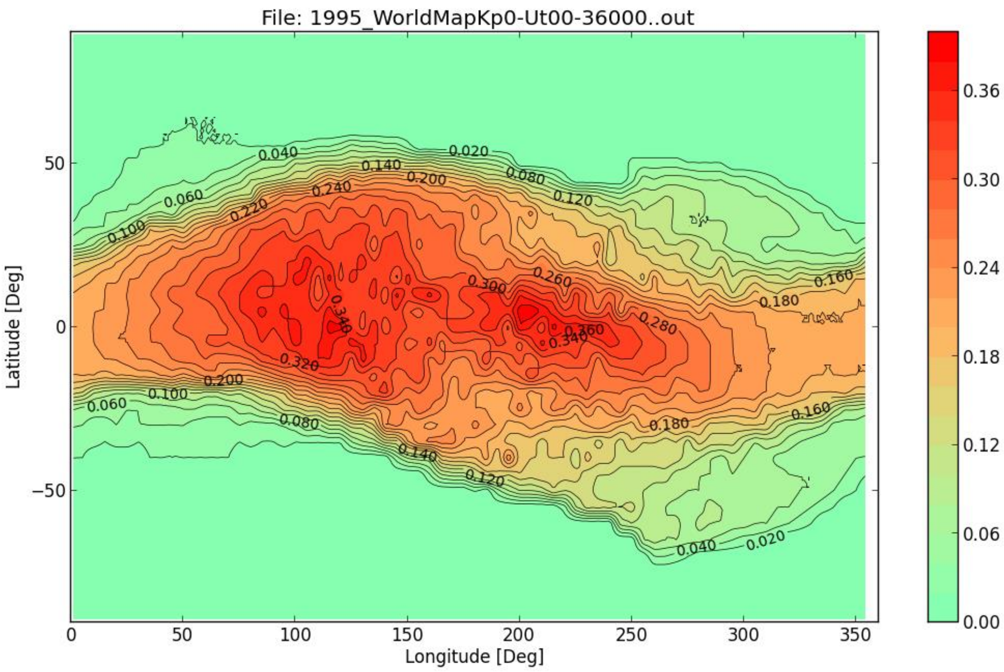

Figure 45: Vertical cut-off (in GV) map at 3600km altitude calculated using MAGNETOCOSMICS for the epoch of 1995.0, UT = 0 hr and kp index of 0.

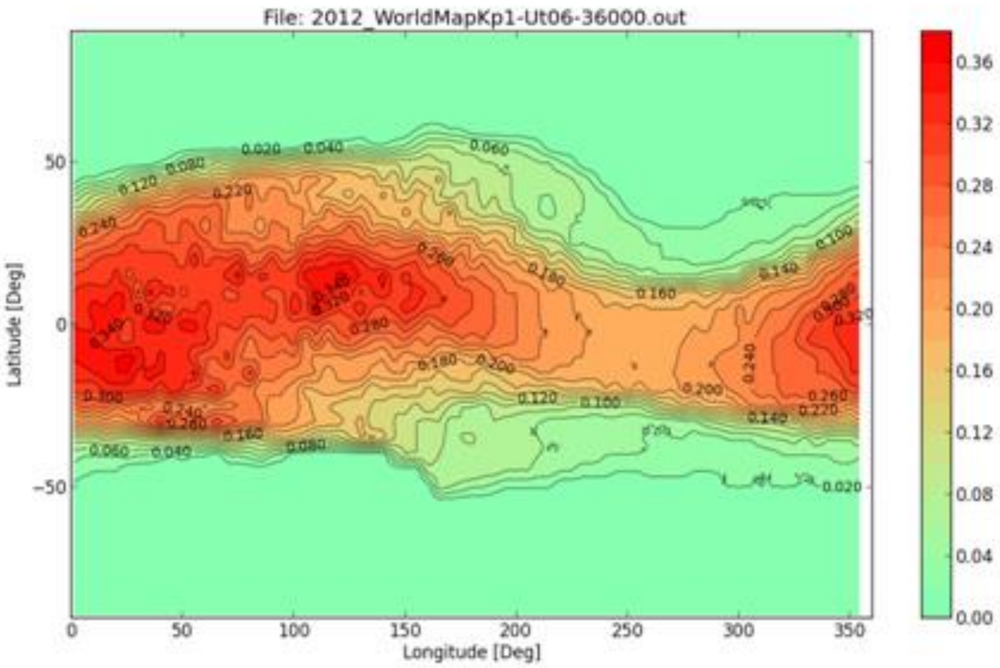

Figure 46: Vertical cut-off (in GV) map at 3600km altitude calculated using MAGNETOCOSMICS for the epoch of 1995.0, UT = 6 hr and kp index of 1.

6.2 Seasonal Effect

The seasonal (day of year) effect to the cut-off rigidity has not been treated explicitly in MSM interpolation algorithms, as the pre-calculated maps are at the beginning of a year at the fixed epochs (1970.0, 1975.0 etc.). However as the interpolation is based on the MacIlwain parameter L, which is calculated including the Day of Year as one of the key input parameters, this implies the MSM calculated cut-offs should in fact reflect the influence of the seasonal effect to the magnetic fields.
In Figure 53 shows the difference of the cut-off between 01/07/1995 and 01/01/1995, as calculated by MSM. Figure 54 is the distribution of this difference.
In Figure 55 shows the difference of the cut-off between 01/07/1995 and 01/01/1995, as calculated by MAGNETOCOSMICS. Figure 56 is the distribution of this difference.
The seasonal effect has been fully accounted for in the MAGNETOCOSMICS results, thus similar plots of the MSM results demonstrate that the MSM code has taken into account the seasonal effect to its rigidity cut-off.

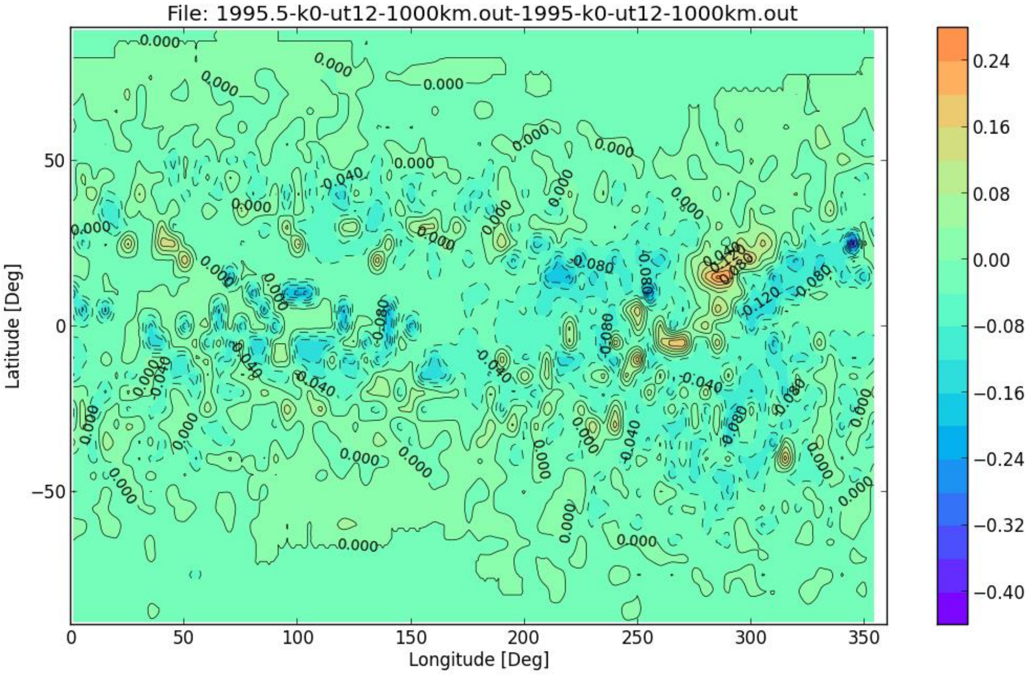

Figure 47: The difference between the cut-off (in GV) maps calculated by MSM for 01/07/1995 and 01/01/1995.

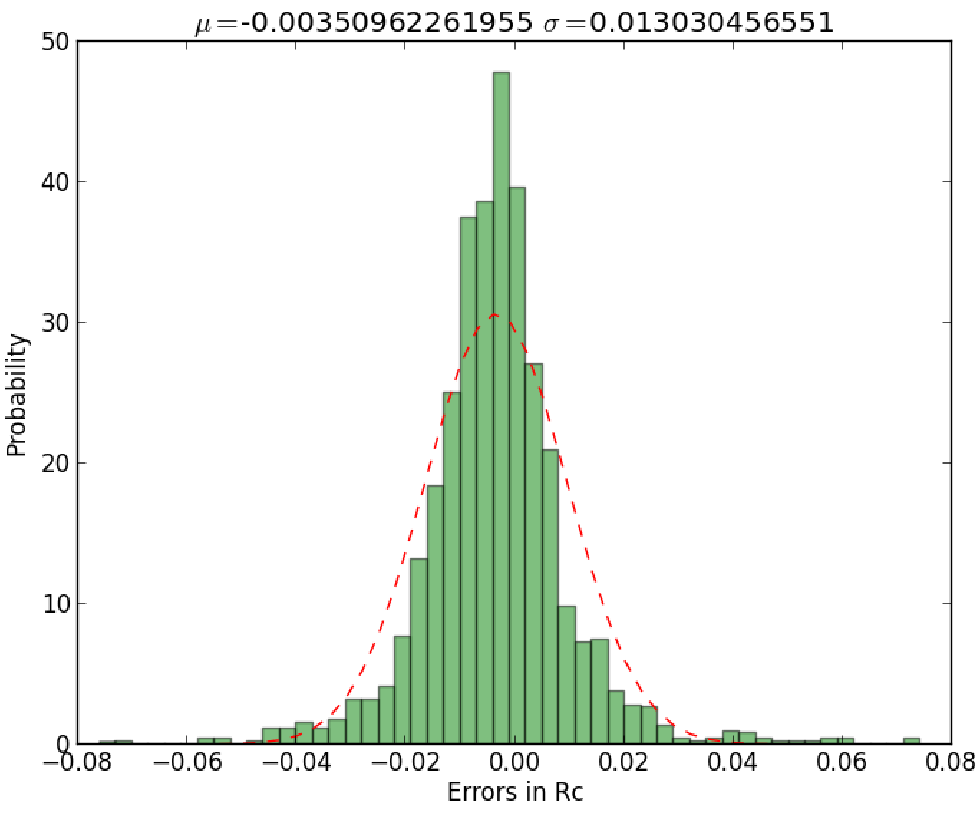

Figure 48: The distribution of the differences in the map shown in figure 52.

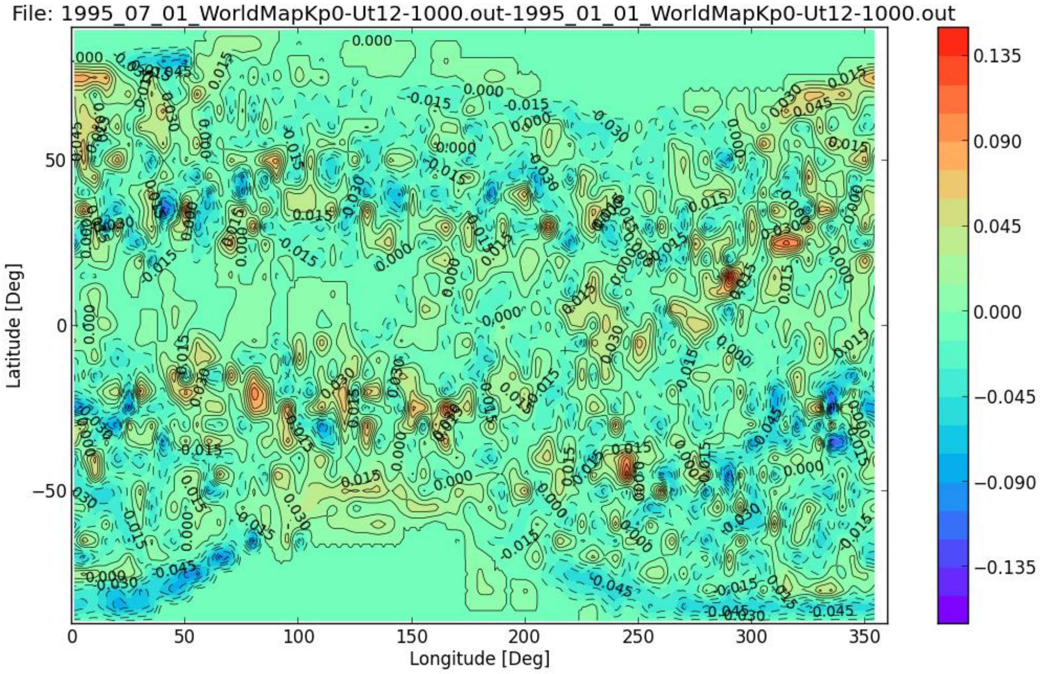

Figure 49: The difference between the cut-off (in GV) maps calculated by MAGNETOCOSMICS for 01/07/1995 and 01/01/1995.

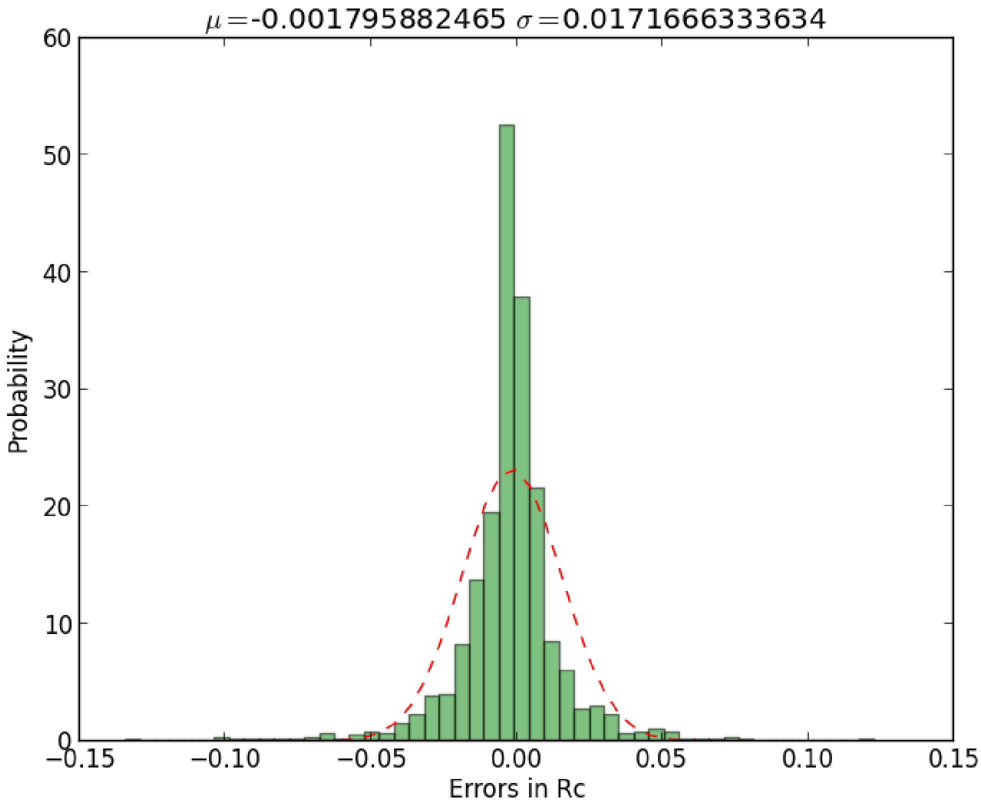

Figure 50: The distribution of the differences in the map shown in figure 54.

6.3 Directional Variations in Cut-off Rigidity

A number of MAGNETOCOSMICS simulations have been carried out to assess the cut-off rigidities as a function of the incident direction of the charged particles.

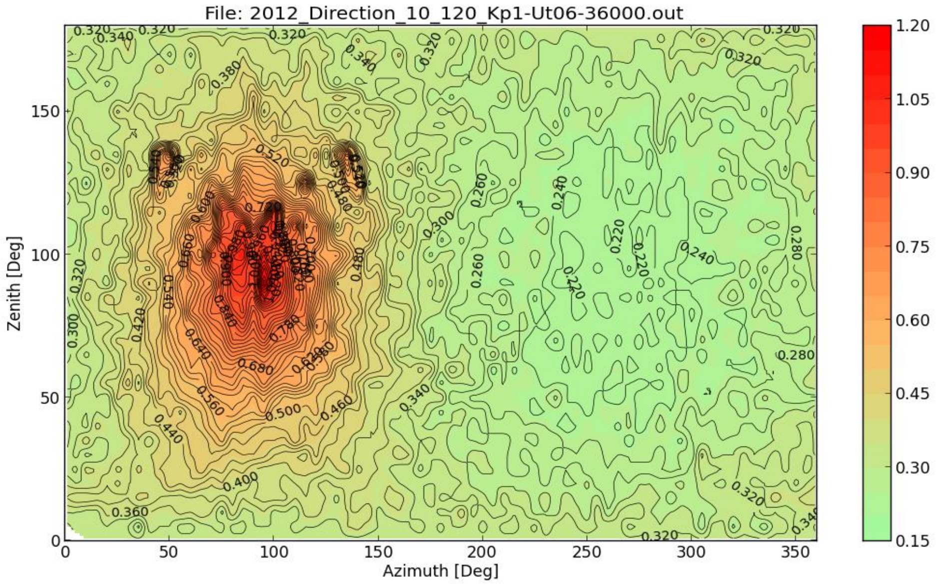

Figure 51: The cut-off rigidity (in GV) as a function of the incident angle of the charged particle. The map is for 01/01/2012 ut=06hr and kp=1. The location is 36000km altitude, 10˚ latitude and 120˚ longitude. The zenith and azimuth angles are local with zenith = 0˚ pointing upwards and azimuth 0˚ is pointing to the magnetic north.

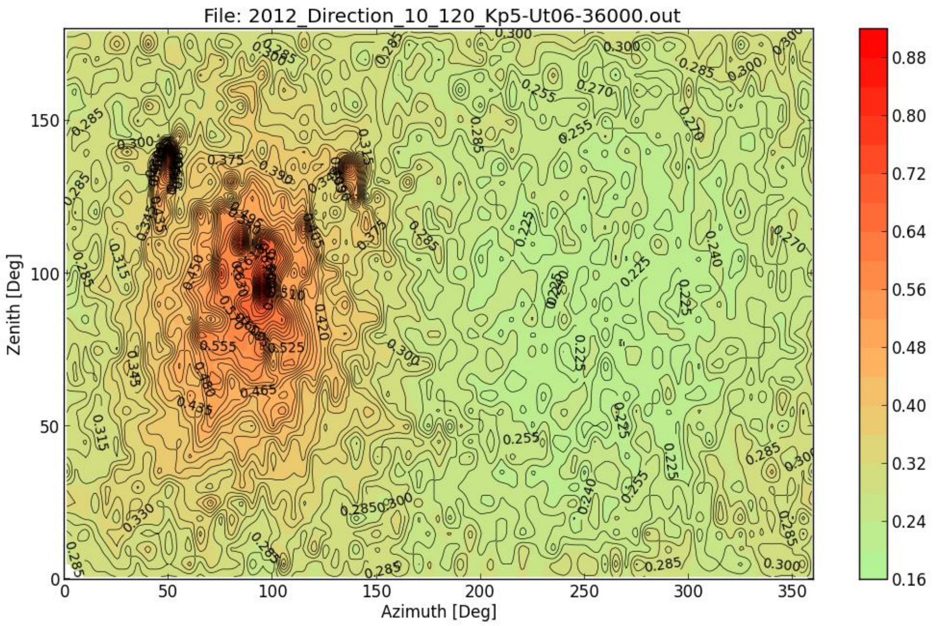

Figure 52: The cut-off rigidity (in GV) as a function of the incident angle of the charged particle. The map is for 01/01/2012 ut=06hr and kp=5. The location is 36000km altitude, 10˚ latitude and 120˚ longitude. The zenith and azimuth angles are local with zenith = 0˚ pointing upwards and azimuth 0˚ is pointing to the magnetic north.

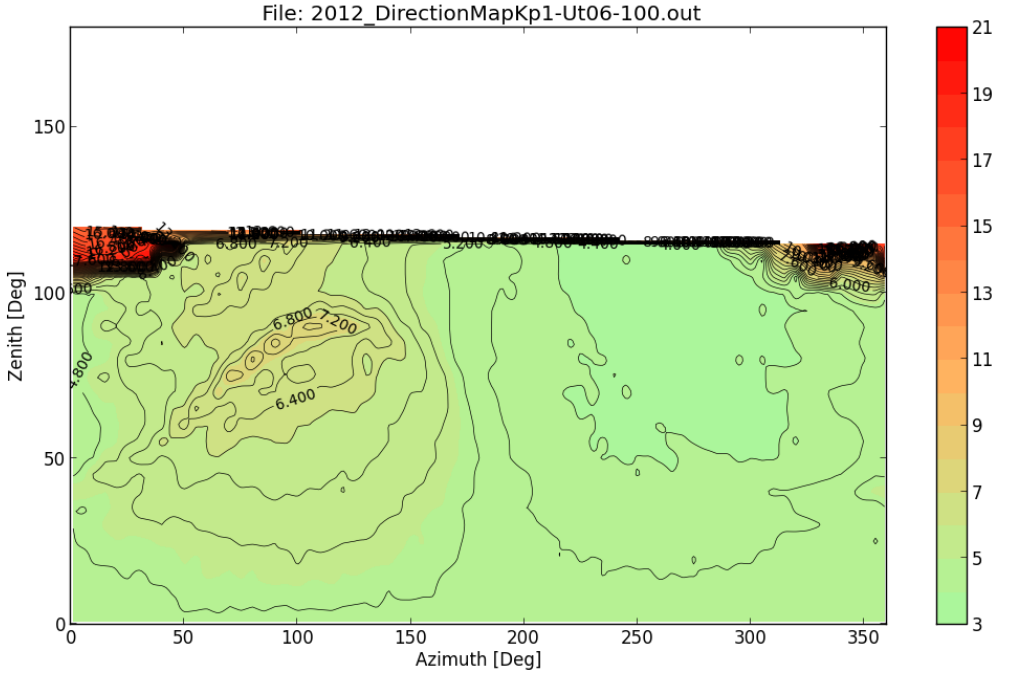

Figure 53: The cut-off rigidity (in GV) as a function of the incident angle of the charged particle. The map is for 01/01/2012 ut=06hr and kp=1. The location is 100km altitude, 0˚ latitude and 0˚ longitude. The zenith and azimuth angles are local with zenith = 0˚ pointing upwards and azimuth 0˚ is pointing to the magnetic north.

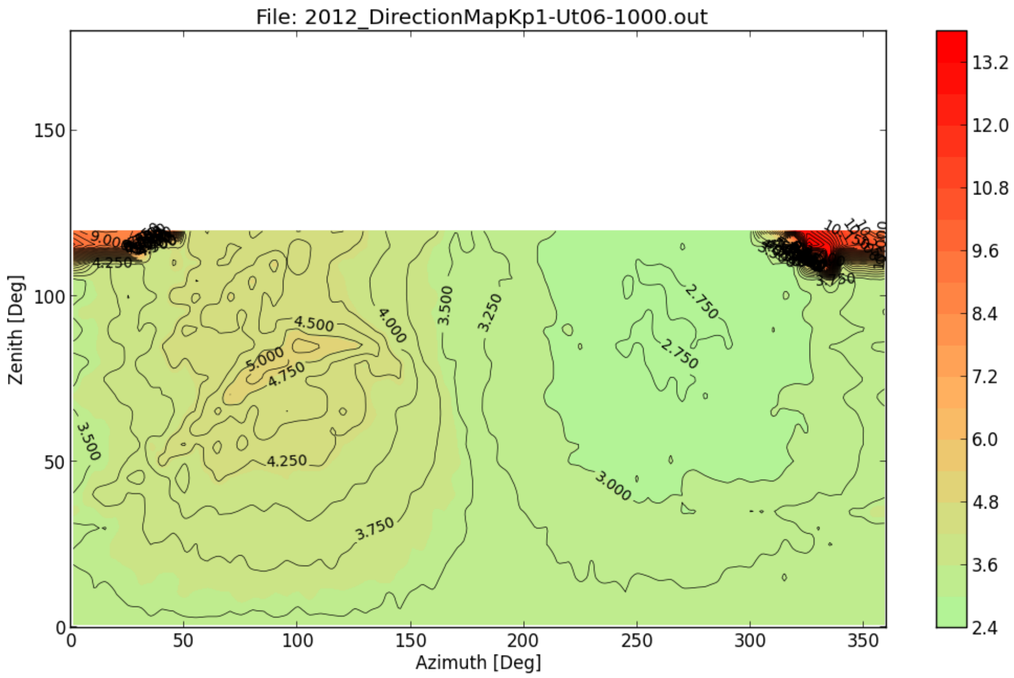

Figure 54: The cut-off rigidity (in GV) as a function of the incident angle of the charged particle. The map is for 01/01/2012 ut=06hr and kp=1. The location is 1000km altitude, 0˚ latitude and 0˚ longitude. The zenith and azimuth angles are local with zenith = 0˚ pointing upwards and azimuth 0˚ is pointing to the magnetic north.

6.4 Epoch, UT and Kp effects on the cut-off rigidities

6.4.1 Epoch

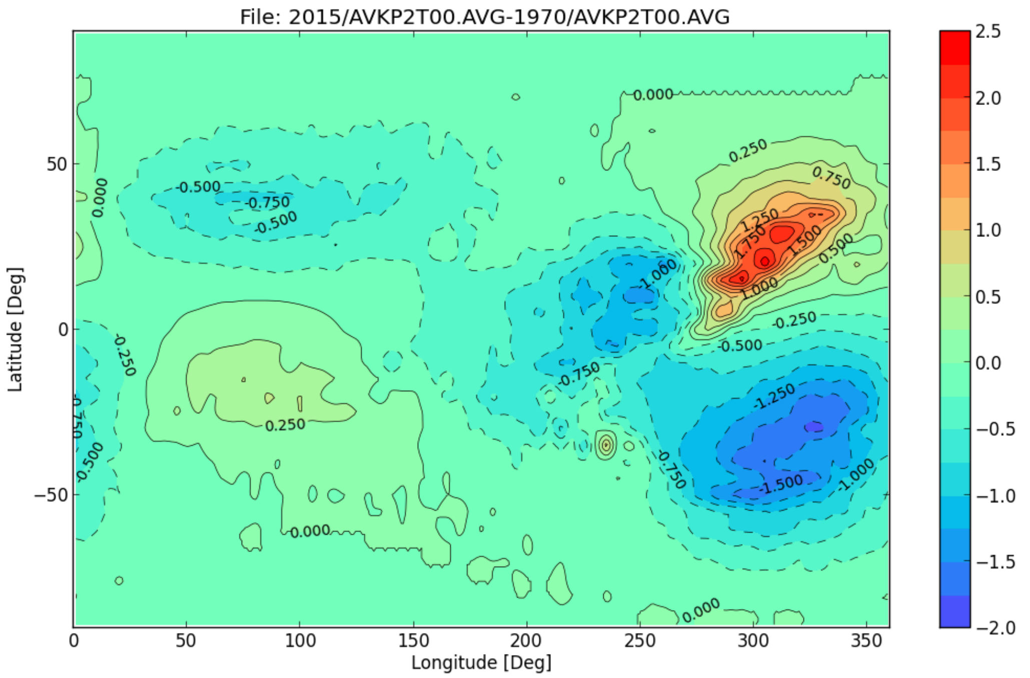

Figure 55: The changes in the vertical cut-off rigidity (in GV) at 450 km altitude between 1970 and 2015. Other conditions are KP =2 and UT = 0 hr.

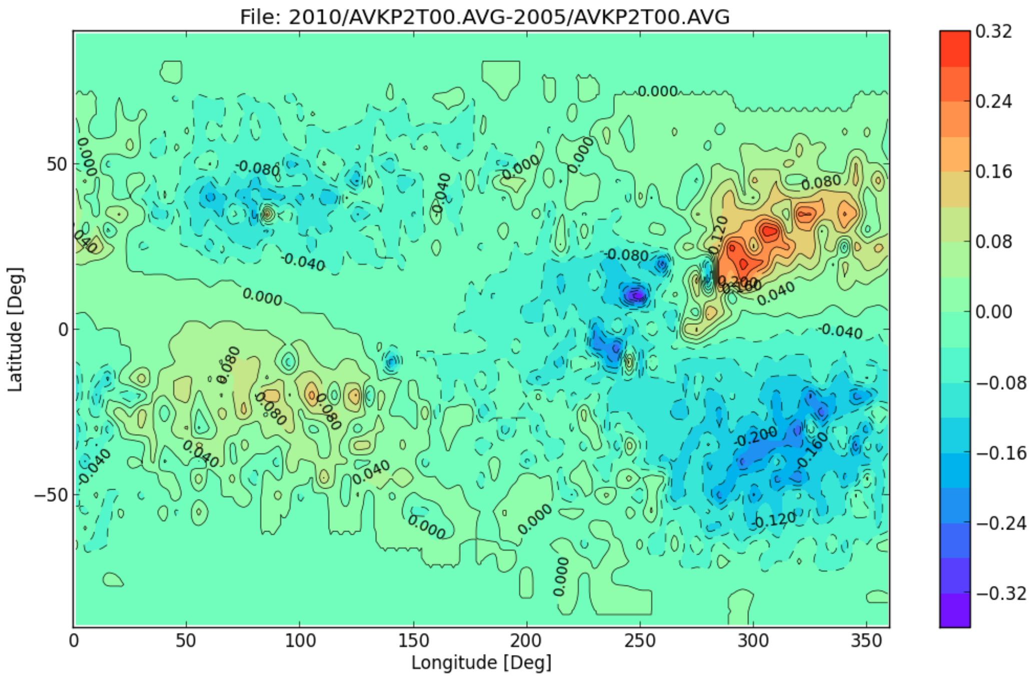

Figure 56: The changes in the vertical cut-off (in GV) rigidity at 450 km altitude between 2005 and 2010. Other conditions are KP =2 and UT = 0 hr.

6.4.2 Kp index

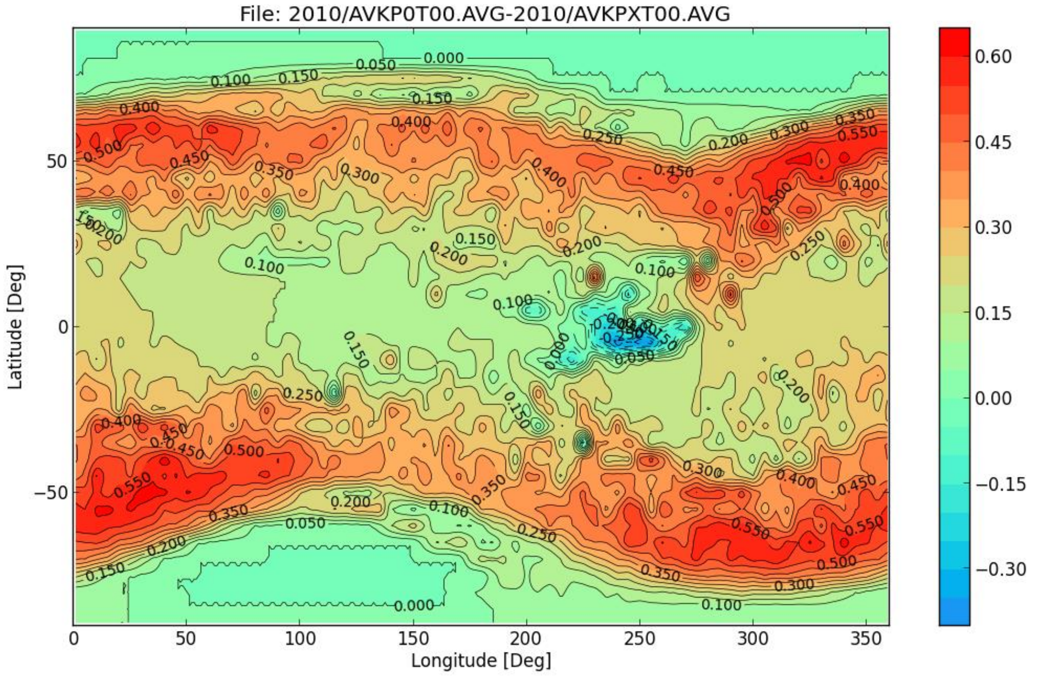

Figure 57: The difference in the vertical cut-off rigidity (in GV) at 450 km altitude between Kp=0 and Kp = 9. Other conditions are Epoch = 2010.0 and UT = 0 hr.

6.4.3 Local Time (UT)

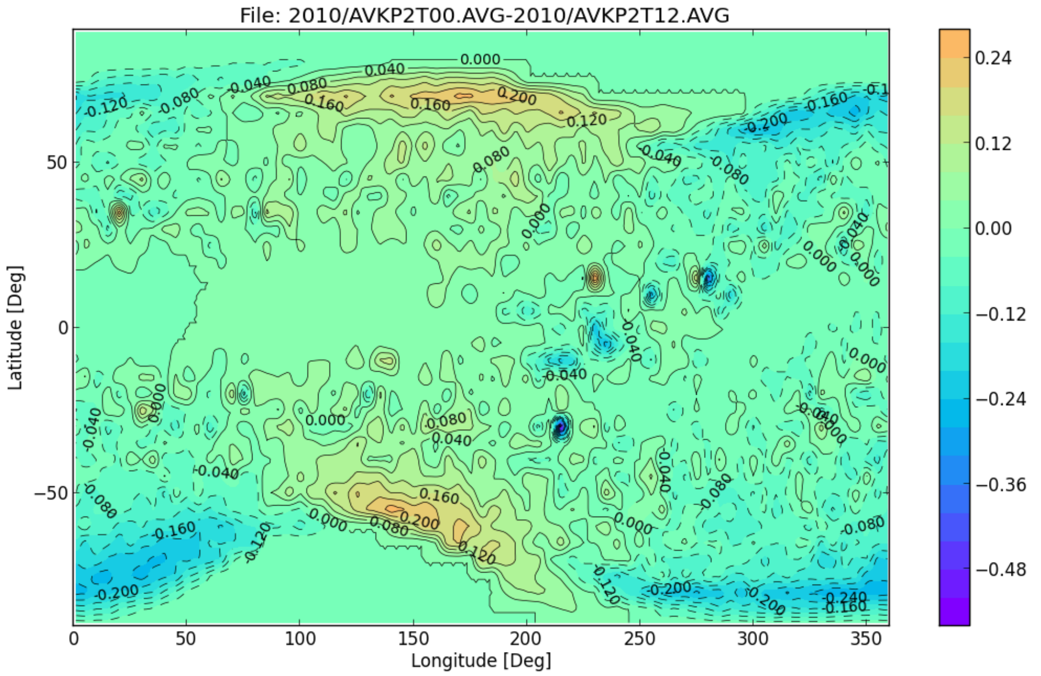

Figure 58: The difference in the vertical cut-off rigidity (in GV) at 450 km altitude between UT=0 and UT = 12 hrs. Other conditions are Epoch = 2010.0 and Kp = 2 .

6.5 Others

1) The quality/statistics of the pre-calculated maps should be improved, in the region around 240 longitude in particular.

2) It seems the MSM tool failed to produce the correct transmission functions in the space around the South Atlantic Anomaly (SSA). This should be investigated further in future studies.

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Initial distribution list

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